Search Effort, Attachment and Labor Market Power

Dominik Wehr Stockholm School of Economics

November 8, 2026

Search Effort and Labor Market Competition

- One in two job seekers is already employed
  → Job-job transitions central to labor market competition

- Search behavior is an integral part of labor mobility
  - ◇ Workers **generate choices through costly effort**

- Yet, little known about search effort and labor market competition

- Kline (Handbook of Labor Economics, 2025)
  However, standard discrete choice formulations of labor supply do not explicitly account for search frictions, which [...] may interact with worker preferences in complex ways.

- Lack of data on on-the-job search:
  1. **How search effort varies across firms**
  2. **Returns to search for workers**

This Paper

- **Analytical model**
  - ◇ Interaction between preference heterogeneity and search frictions
  - ◇ Derive implications for labor supply elasticity

- **Reduced-form evidence**
  - ◇ Labor force survey linked to administrative data
  - ◇ Observe **search, motives, and subsequent mobility**
  - ◇ Measure **variation in on-the-job search across firms**
  - ◇ Provide **reduced-form decomposition of returns** to search
  - ◇ Characterize **persistent preference heterogeneity**

- **Quantitative dynamic discrete-search model**
  - ◇ **Structural search elasticity** identified using search and mobility data
  - ◇ Quantify contribution to labor supply elasticity

Preview of Results

- **Three empirical facts about search**
  - ◇ Search effort decreases with firm pay
    $\rightarrow$ Reduced-form: 1% higher pay $\rightarrow$ 0.33pp lower search
  - ◇ Search driven by idiosyncratic, non-pecuniary motives
    $\rightarrow$ Downward mobility across firms largely voluntary
  - ◇ Match-specific amenities evolve within employment spells

- **Model: implications for firm's labor supply**
  - ◇ **Endogenous search:** amplifies retention elasticities
    – Retention: 1.62 $\rightarrow$ 2.95 $(\times 1.8)$
    – Employment: 3.71 $\rightarrow$ 5.03 $(\times 1.4)$
    – Asymmetry between recruitment and retention
  - ◇ **Heterogeneity in propensity to search matters**
    – Attenuates amplification 44% below representative benchmark

$\Rightarrow$ **Endogenous choice sets strongly shape firm's labor supply**

Literature

- **Firm labor supply and wage-setting**
  - ◇ **Kline (2025);** Manning (2021); Card (2022); Berger et al. (2022, 2024); Dal Bó et al. (2013); Falch (2017); Belot et al. (2022); Datta (2023); Roussille and Scuderi (2026)
  - ◇ **Here:** Contribution of endogenous search to labor supply elasticity
- **Job amenities, compensating differentials, and sorting**
  - ◇ Fujita (2010); Mas and Pallais (2017); Wiswall and Zafar (2018); Maestas et al. (2023); Sorkin (2018); Hall and Mueller (2018); Taber and Vejlin (2020); Lamadon et al. (2022); Morchio and Moser (2026); Jarosch (2023); Humlum et al. (2025); Caldwell et al. (2025, 2026)
  - ◇ **Here:** Reduced-form decomposition of reasons for search across firms
- **Endogenous on-the-job search**
  - ◇ Postel-Vinay and Robin (2004); Christensen et al. (2005); Nagypál (2007); Lise (2013); Bagger and Lentz (2019); Eeckhout and Lindenlaub (2019); Faberman et al. (2022); Simmons (2024); Afrouzi et al. (2026); Pilossoph et al. (2026); Kaas et al. (2026); Faberman et al. (2026)
  - ◇ **Here:** Develop dynamic discrete choice model with wage-posting
    → Empirically estimated on firm data + longitudinal search dynamics

Labor Supply Conditional on Offers

- A worker chooses between her current job and an outside offer

$$\mathcal{C} = \{f, g\}, \qquad j^* = \arg\max_{j \in \mathcal{C}} \{V_j + \eta_j\}$$

- Independent Gumbel taste shocks imply

$$p_f \equiv \Pr(f \mid \mathcal{C}) = \frac{\exp(V_f/\sigma)}{\exp(V_f/\sigma) + \exp(V_g/\sigma)}$$

$V_j = V(w_j)$; taste shocks are redrawn at each offer arrival.

- The same choice has two interpretations:
  - ◇ **Recruitment:** a worker at g accepts an offer from f
  - ◇ **Retention:** a worker at f rejects an offer from g

$\Rightarrow$ **Higher pay increases firm's recruitment and retention** conditional on choice set

How Are Offers Generated?

- **Random search with wage posting**
  - ◇ Identical workers; direct employer-to-employer mobility
  - ◇ Outside offers are drawn from a fixed wage distribution F
  - → Workers **cannot direct search** to high-pay firms
- Workers choose **search effort** $s \geqslant 0$
- Effort determines the **arrival rate of outside offers**

$$\lambda(s) = \underbrace{\lambda_0}_{\text{Unsolicited offers}} + \underbrace{\lambda_1 s}_{\text{Offers generated by search}}$$

- Search is **costly**

$$c(s) = \frac{\kappa}{2} s^2, \qquad \kappa > 0$$

⇒ **Endogenous formation of the choice set**

How does search vary with firm quality V?

- Let $\Omega(V)$ denote the **expected gain from an additional outside offer**
- **Optimal search** equates marginal cost and expected benefit

$$\kappa s(V) = \lambda_1 \Omega(V)$$

- Let $\mathcal{A}(V)$ denote **acceptance rate**, conditional on receiving outside offer
- Search effort declines with offered utility

$$\frac{\partial s(V)}{\partial V} = -\frac{\lambda_1 \mathcal{A}(V)}{\kappa} < 0$$

- High acceptance rate: many outside offers are suitable $\rightarrow$ steepens response

$\Rightarrow$ **Higher compensation reduces exposure to competing offers**

Endogenous Search Strengthens Retention

Raise pay w at one small employer; keep the rest of the market fixed.

$$Q(w) = \lambda(s(w))\,\mathcal{A}(V(w)), \qquad N(w) = \frac{R(w)}{Q(w)}$$

Q: **quit** rate per worker; R:**hiring rate**; N: stationary employment.

Labor supply elasticity wrt wage $\epsilon^N = \epsilon^R - \epsilon^Q$.

Frozen search (C)

Search fixed at its baseline optimum.
Acceptance responds to pay.

Endogenous search (E)

Search and acceptance respond to pay.
Offer arrival also falls.

Additional contribution to labor supply elasticity

$$\epsilon^{N,E} - \epsilon^{N,C} = \underbrace{\frac{\lambda_1 s}{\lambda_0 + \lambda_1 s}}_{\text{Active-offer share}} \times \underbrace{(-\epsilon^s)}_{\text{Search response to pay}} > 0$$

$\Rightarrow$ **Higher pay retains workers by reducing exposure to competing offers.**

Search and Preferences Interact: A Symmetric Benchmark

- Assume a degenerate outside wage distribution at $\bar{w}$: $P(V(\bar{w}), V(\bar{w})) = \frac{1}{2}$
- With offer arrivals held fixed, recruitment and retention contribute equally:

$$\epsilon^{N,C} = 2\epsilon^{R}$$

- With endogenous search and quadratic costs:

$$\epsilon^{N,E} = \left(2 + \frac{\sigma \lambda_1^2}{\kappa \lambda(s)}\right) \epsilon^{R}$$

- Amplification depends on **interaction** of
  - ◇ **Preference heterogeneity σ**
  - ◇ **Search costs κ** and **offer process $\lambda(\cdot)$**
  - ◇ Prevailing **search effort s**

$\Rightarrow$ Search amplifies labor supply beyond recruitment and acceptance-based retention

Match Heterogeneity and the Search Channel

Assumptions

- Allow for **persistent preference heterogeneity** $\mathfrak{q}$: $V(w) \longrightarrow V(w, \mathfrak{q})$

Match Heterogeneity and the Search Channel

Assumptions

- Allow for **persistent preference heterogeneity** q: $V(w) \longrightarrow V(w, q)$
- Search and acceptance now differ across workers at the same employer

$$Q(w, q) = \lambda(s(w, q))\mathcal{A}(V(w, q))$$

Match Heterogeneity and the Search Channel

Holding workforce composition fixed

$$\mathcal{C}_{\text{quit}} = \theta^{Q} \Bigg[\underbrace{-\varepsilon^{\bar{s}}}_{\text{Search elasticity}} + \underbrace{\frac{\text{Cov}_s(\mathcal{A}, -\varepsilon^s)}{\mathbb{E}_s[\mathcal{A}]}}_{\text{Acceptance selection}} \Bigg]$$

- **Active hire share:** θ^{Q} Share of quits via active search
- **Search elasticity:** $\varepsilon^{\bar{s}}$ behavior response of search to pay
- **Acceptance selection:** Covariance between acceptance rate $\mathcal{A}(q)$ and elasticity
  $\rightarrow$ How does search response vary across workers?

$\Rightarrow$ **The same reduction in search need not prevent the same number of quits.**

Match Heterogeneity and the Search Channel

Allowing the workforce to adjust

$$\varepsilon^{N,E} - \varepsilon^{N,C} = \theta^{Q} \Bigg[\underbrace{-\varepsilon^{\bar{s}}}_{\text{Search elasticity}} + \underbrace{\frac{\mathrm{Cov}_s(\mathcal{A}, -\varepsilon^s)}{\mathbb{E}_s[\mathcal{A}]}}_{\text{Acceptance selection}} + \underbrace{\mathbb{E}_{s\mathcal{A}}\left[(-\varepsilon^s)(\bar{Q}T(q) - 1)\right]}_{\text{Workforce adjustment}} \Bigg]$$

- **Active hire share:** θ^{Q} Share of quits via active search
- **Search elasticity:** $\varepsilon^{\bar{s}}$ behavior response of search to pay
- **Acceptance selection:** Covariance between acceptance rate $\mathcal{A}(q)$ and elasticity
  $\rightarrow$ How does search response vary across workers?
- **Workforce adjustment:** $T(q)$ is expected remaining tenure
  $\rightarrow$ How does composition of workforce change due to search response?

$\Rightarrow$ **Avoided quits preserve different amounts of future employment.**

Linked survey – administrative data for Sweden

- **Labor Force Survey** (LFS, 2005 – 2020)
  - ◇ **Representative**, **large** sample: 0.4% of the population
  - ◇ **Longitudinal** – 8 consecutive quarters
  - ◇ Self-reported information on **on-the-job search**
    1. Actively Looked for another job in the past four weeks?
    2. Number of methods [Proxy for intensity]
    3. **Primary reason for on-the-job search**

⇒ **Search dynamics** and **workers' preferences**

- Linked to economy-wide data on firms
  - ◇ **Matched Employer-Employee Data** (1985 – 2020)
  - ◇ **Firm Balance Sheets** (2005 – 2018)
  - → Measure firm pay premia and labor productivity

⇒ Variation in **on-the-job search across firms**

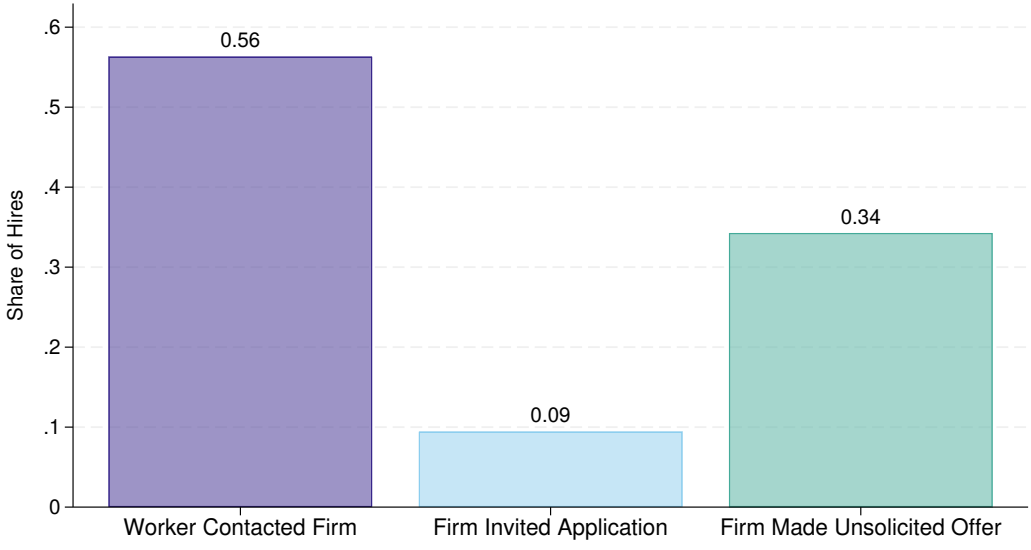
Sample and Institutional Setting

- **Collective agreements govern wage-setting**
  - ◇ Individual pay tied to objective criteria: ability of worker + performance of firm
  - ◇ Legal institutions and unions constrain wage dispersion
  - → Supports **wage-posting** framework

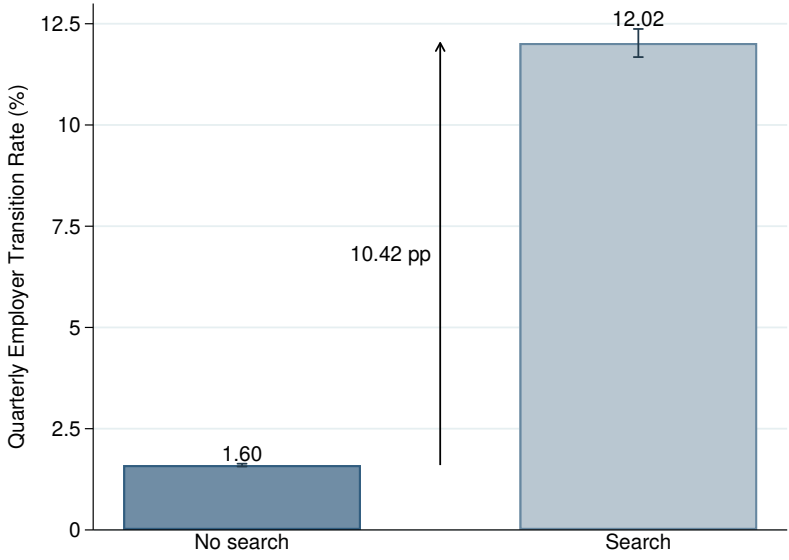
- **Duality in employment protection**
  - ◇ Open-ended jobs protected – generous advance notice
  - ◇ Flexible fixed-term contracts (40 percent of vacancies)
  - ◇ Probationary contracts face limited dismissal protection
  - → **Mobility may entail loss of employment protection**

- **Sample: private-sector employment in Sweden**
  - ◇ **Private-sector firms** only
  - ◇ Exclude industries with non-standard contracts
  - ◇ Require valid value-added information
  - ◇ Restrict to prime-age workers with labor force attachment

Recruitment Channels: Most hires require worker initiative



Searchers More Likely To Switch Employers



Empirical Strategy: How to Measure Firm Quality

1. **Firm pay premia: pay for a given worker**

$$\ln w_{it} = \alpha_i + \psi_{J(i,t)} + X'_{it}\beta + \varepsilon_{it}$$

   - ◇ Separate **firm pay policy** from worker composition
     Abowd et al. (1999)

2. **Value added per worker: labor productivity**
   - ◇ Complementary ranking constructed **without using wages**
   - ◇ Pay can compensate for **less attractive non-pay attributes**
   - → Project firm pay onto value-added to isolate "vertical" variation

- **Robustness: revealed-preference ranking**
  - ◇ Rank firms by the **share of hires poached from other employers**
    Bagger and Lentz (2019)

Empirical Strategy: Measuring the Search Elasticity

- **Relate search to the employer's pay premium**

$$s_{it} = \alpha + \beta_s \widehat{\psi}_{J(i,t)} + X_{it}'\gamma + u_{it}$$

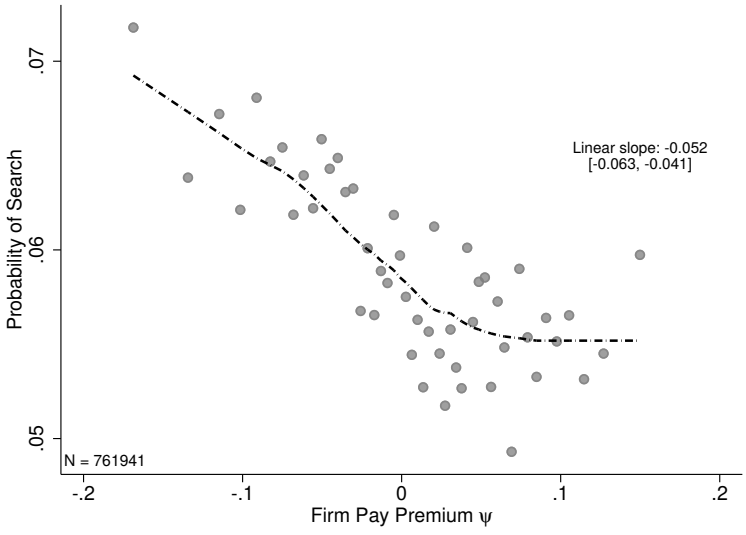
s_{it}: active-search indicator; $\widehat{\psi}_{J(i,t)}$: employer's estimated log-pay premium

- **Identification requires**

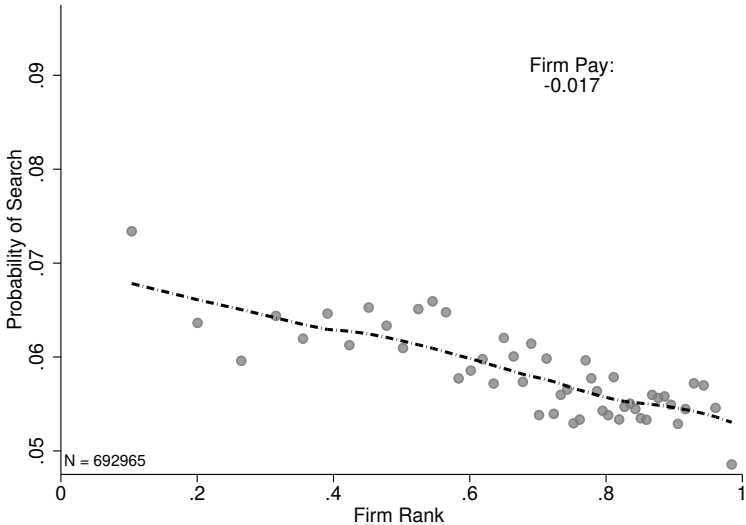
$$\mathrm{Cov}(\widetilde{\psi}_{it}, u_{it}) = 0$$

- Threats to identification
  - ◇ **Compensating differentials**: pay compensates for bad non-pay attributes
    → Rely on value-added to identify utility rank
  - ◇ **Match-value selection**: workers sort on match-specific valuation
    → Panel variation + structural model

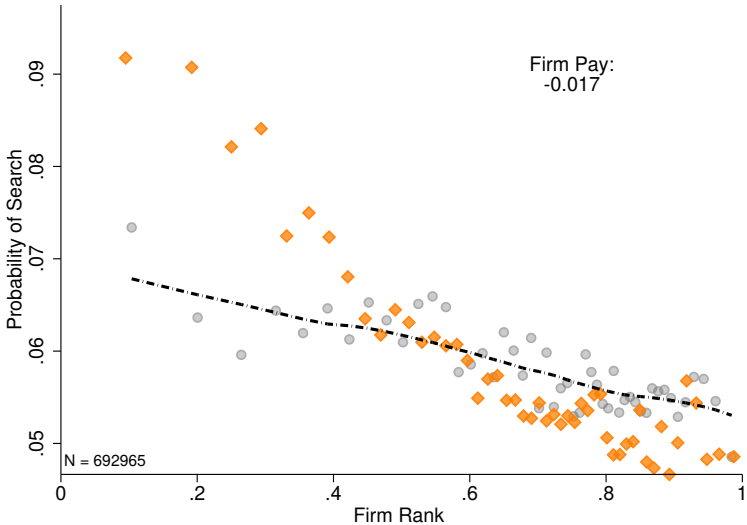
Fact II: Search declines modestly in firm pay



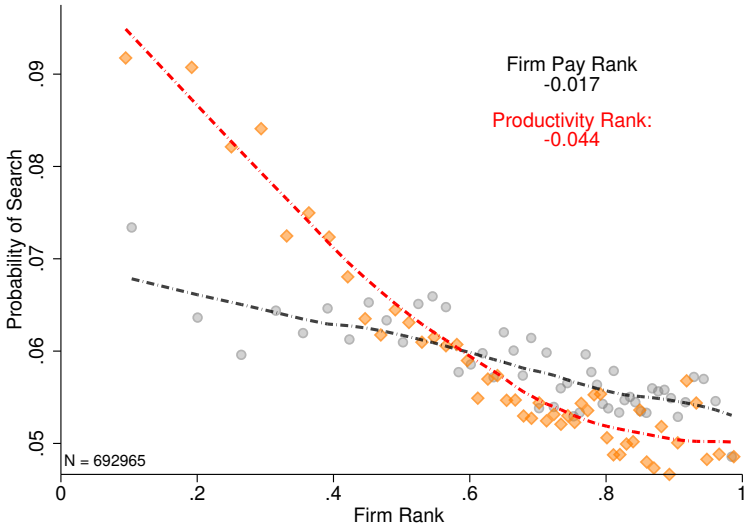
Firm Pay Rank



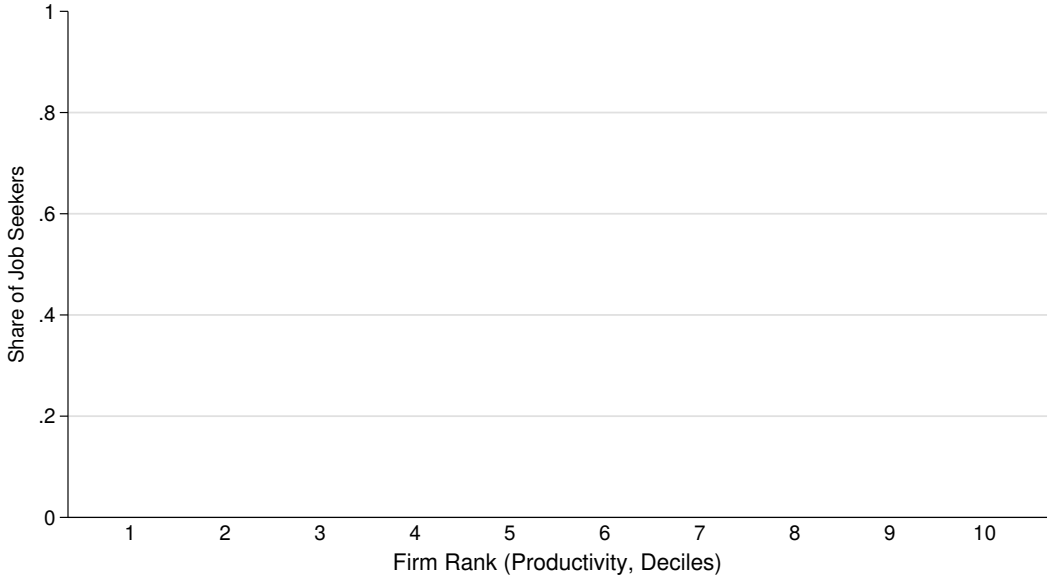
Productivity Rank



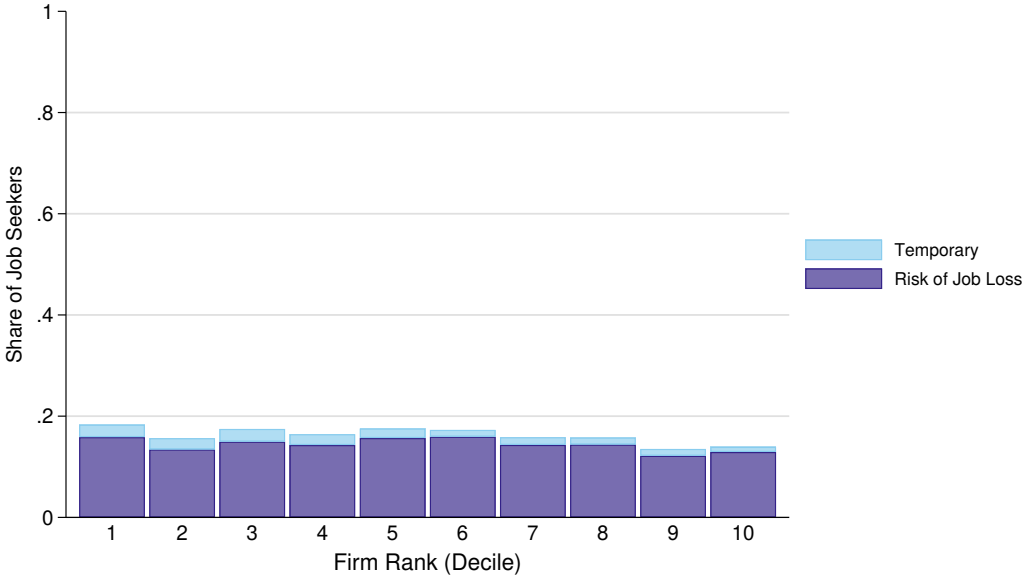
Search gradient significantly steeper in productivity



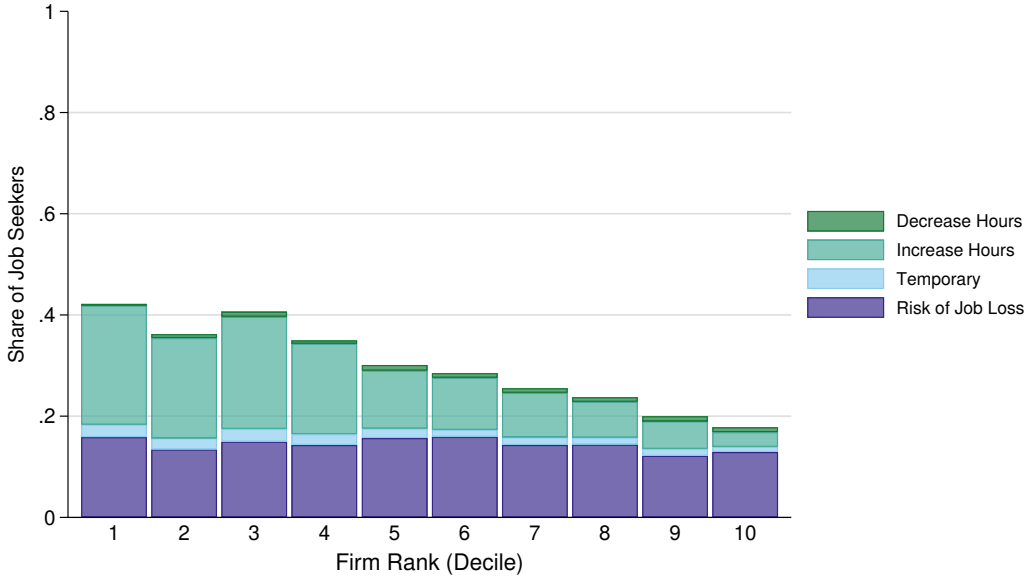
Composition of Motives Across Job Ladder



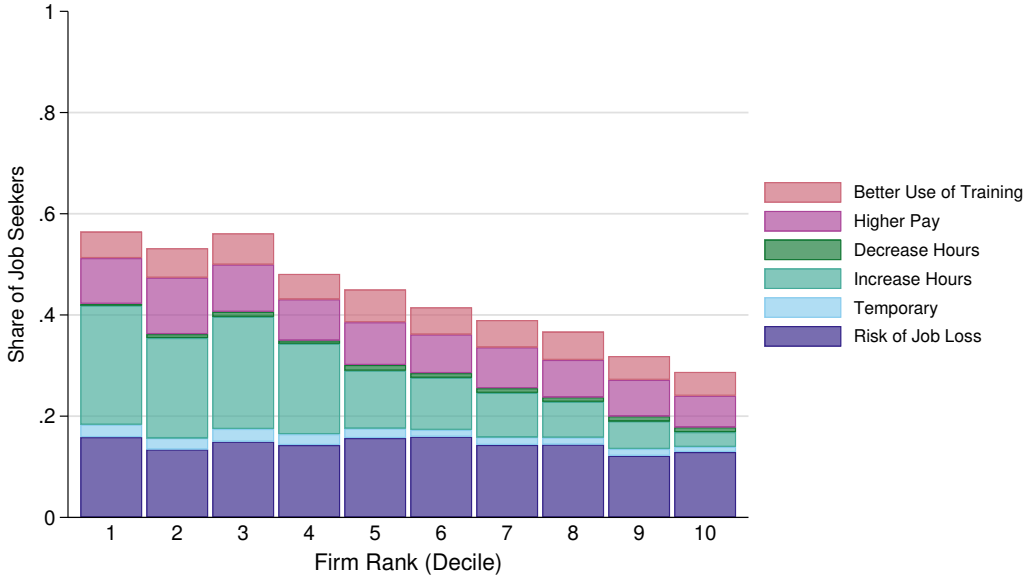
Risk of Job Loss Important at Lower Rungs



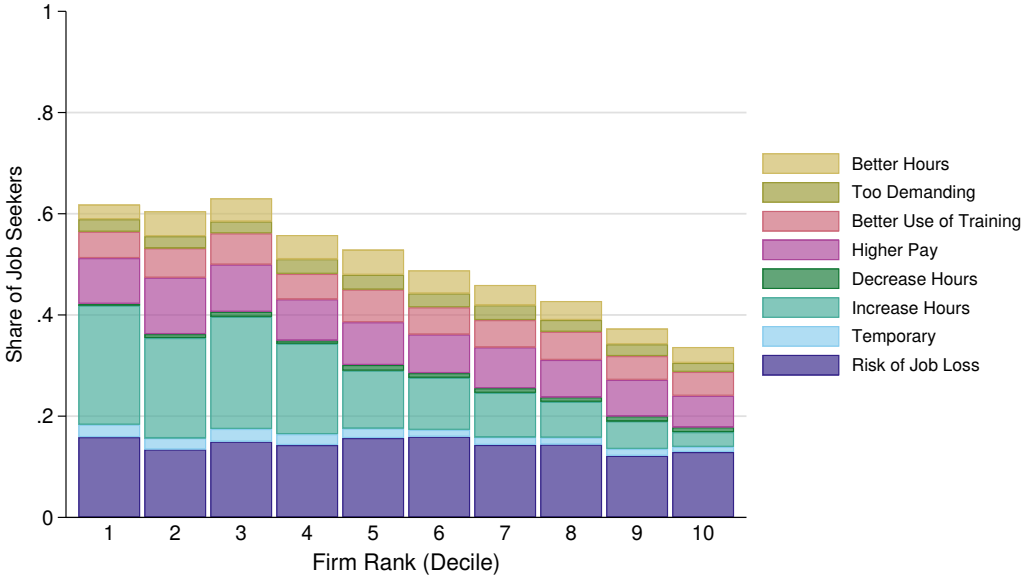
Underemployment Very Important at Lower Rungs



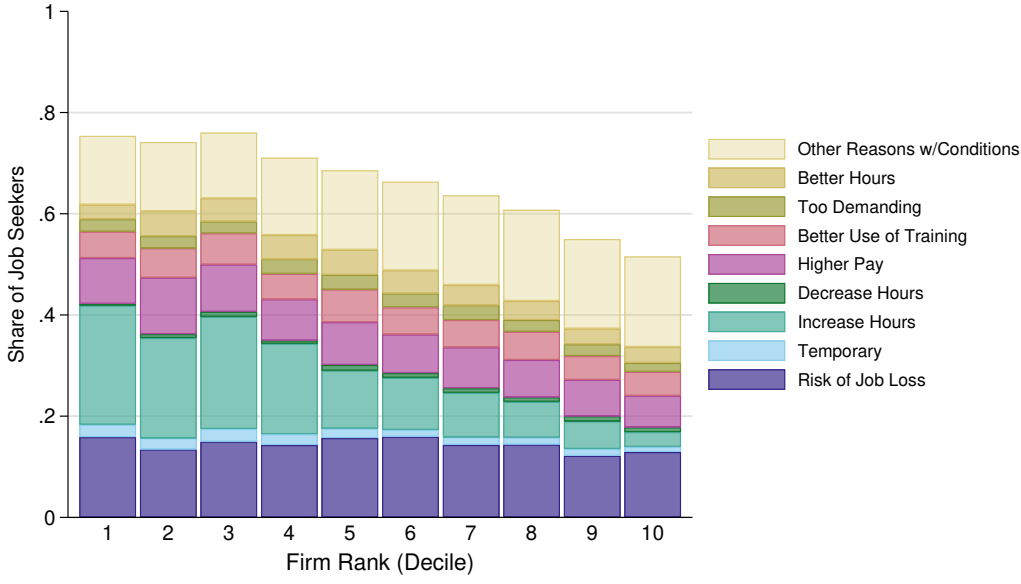
Pay + Education Mismatch Fairly Constant Share



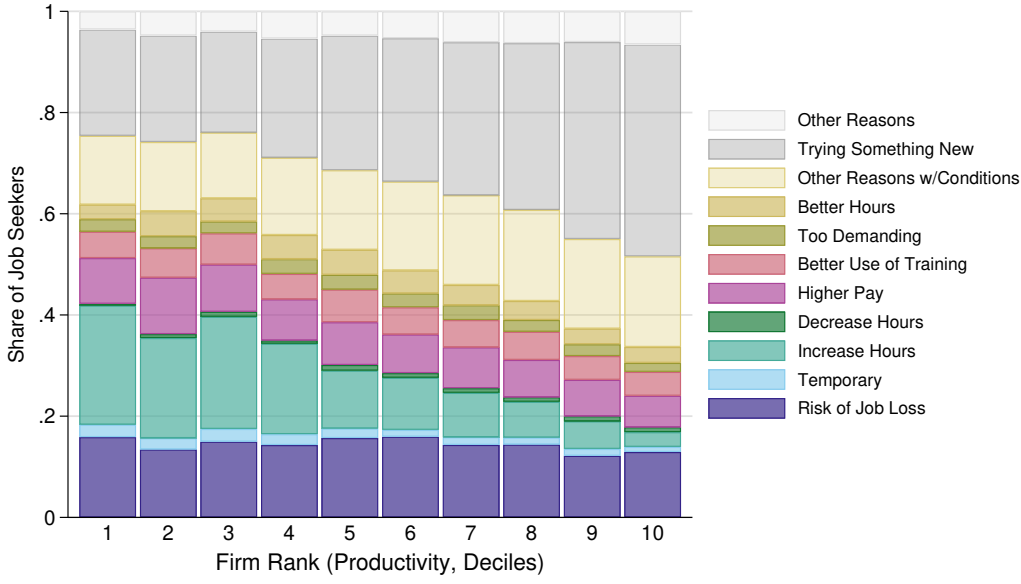
Work Conditions



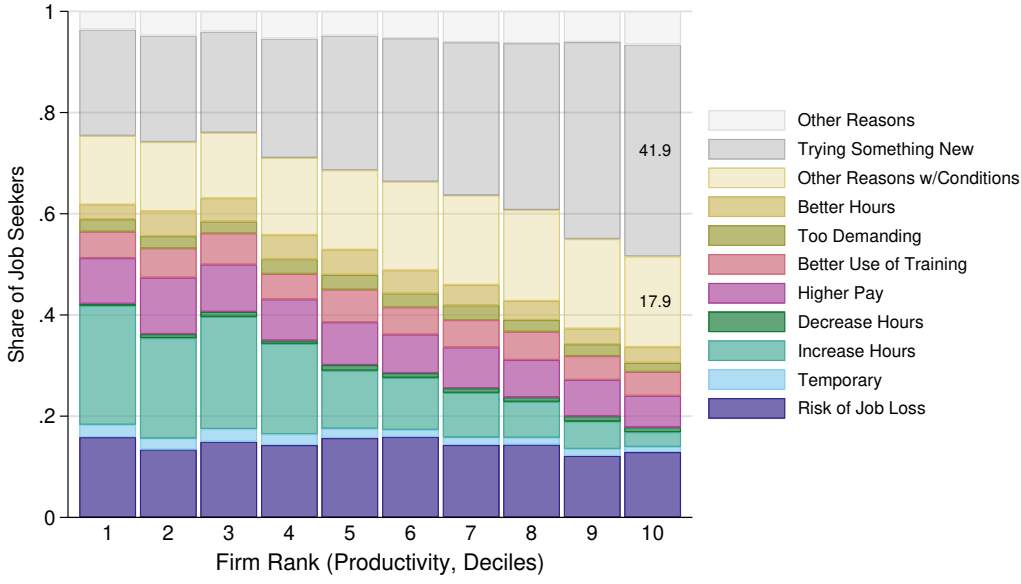
Mixed Work Conditions (e.g. Commute) Increasing



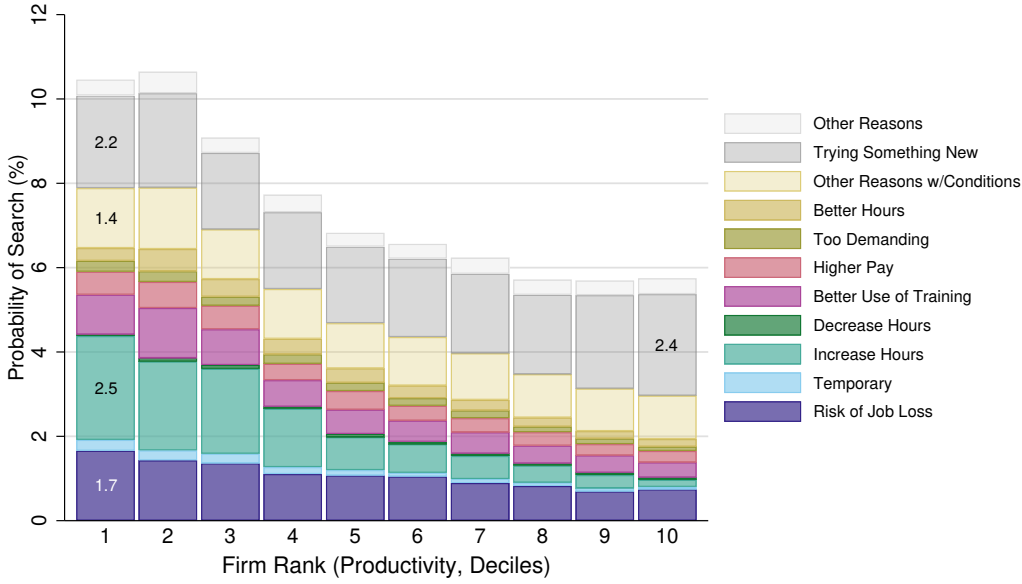
Trying Something New Important at Top



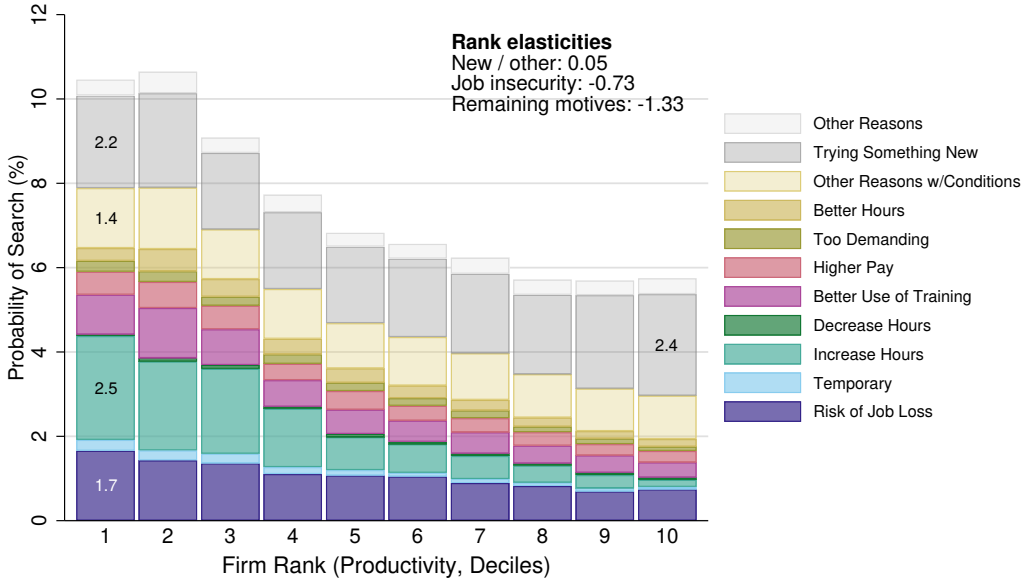
Trying Something New Important at Top



Preferences and Job Loss Account for Intercept



Rank Elasticities Heterogeneous across Motives



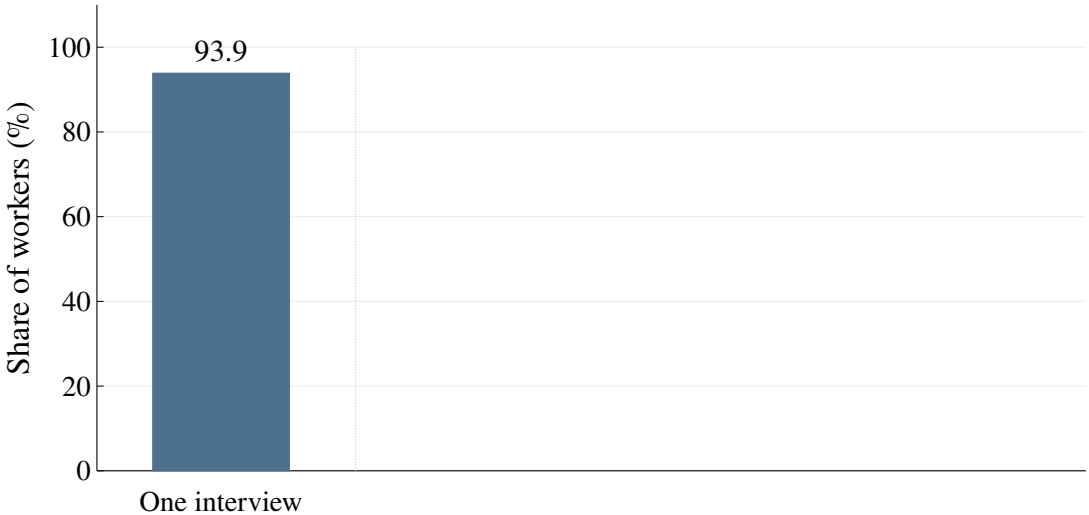
Taking Stock

- **Takeaway**: **Search at top-ranked firms largely voluntary**
  - ◇ Support for **preference heterogeneity**
  - ◇ as opposed to hidden layoffs

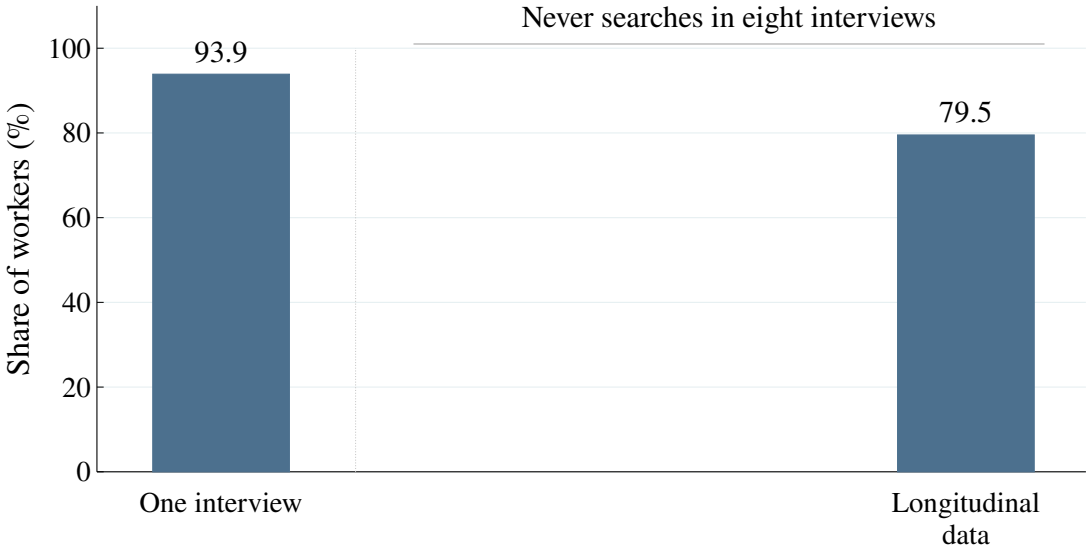
Taking Stock

- **Takeaway**: **Search at top-ranked firms largely voluntary**
  - Support for **preference heterogeneity**
  - as opposed to hidden layoffs
- **Robustness**
  - **Renegotiation:** Limited within-spell improvements following search
  - **Market Definition:** Municipality, industry or flow-based market
  - **Measurement error:** Bias-correction, restrict to large firms
  - **Employer definition:** Firm versus plant
  - **Intensive margin:** Search methods and duration
  - **Revealed preference:** Poaching rank
  - **Motives:** Transition-yield weighting
  - **Subsequent moves:** Motives predict direction of realized moves

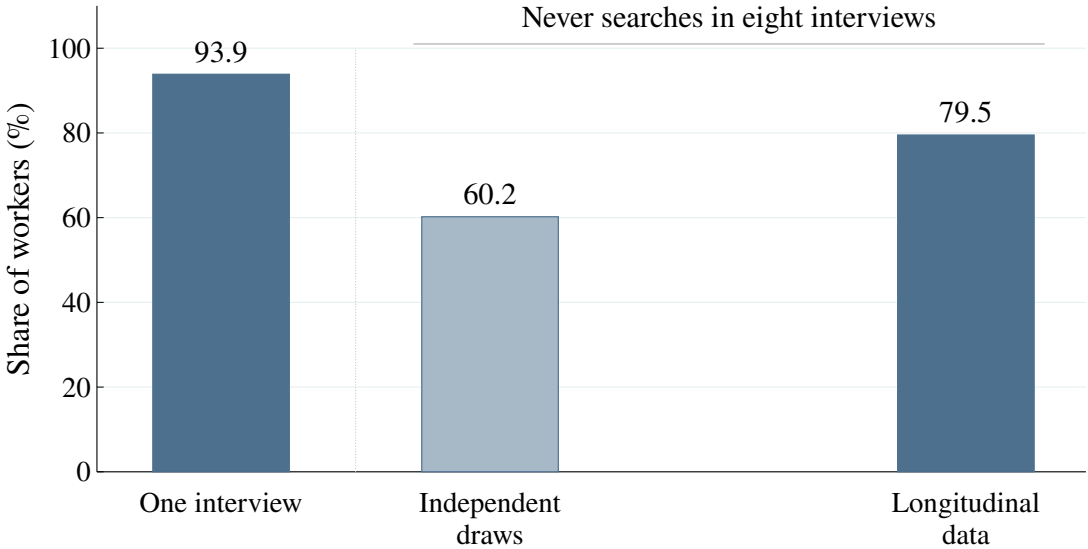
Following Workers Over Time: Search is highly concentrated



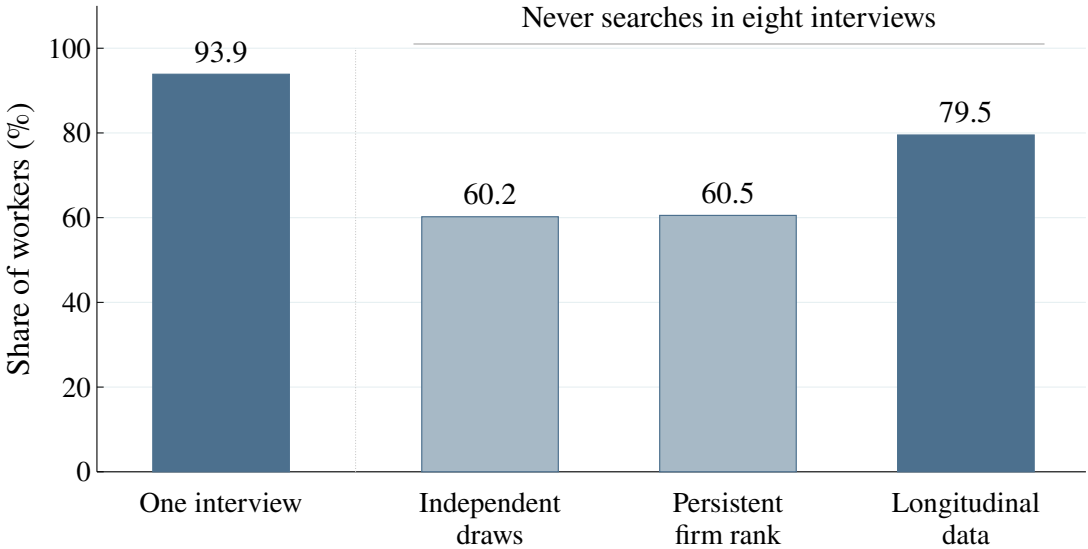
Following Workers Over Time: Search is highly concentrated



Following Workers Over Time: Search is highly concentrated



Following Workers Over Time: Search is highly concentrated

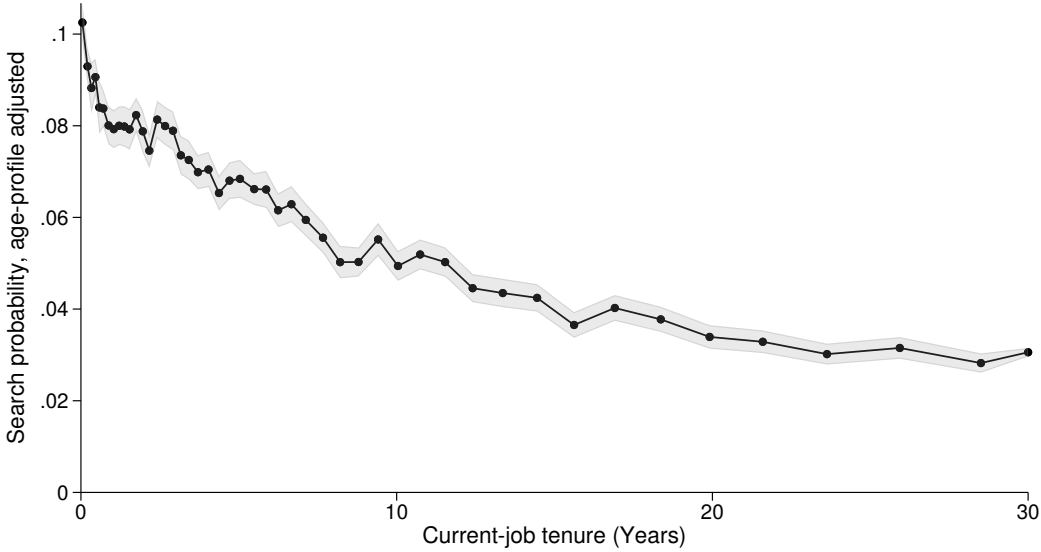


Preference Heterogeneity, Sorting and Dynamics

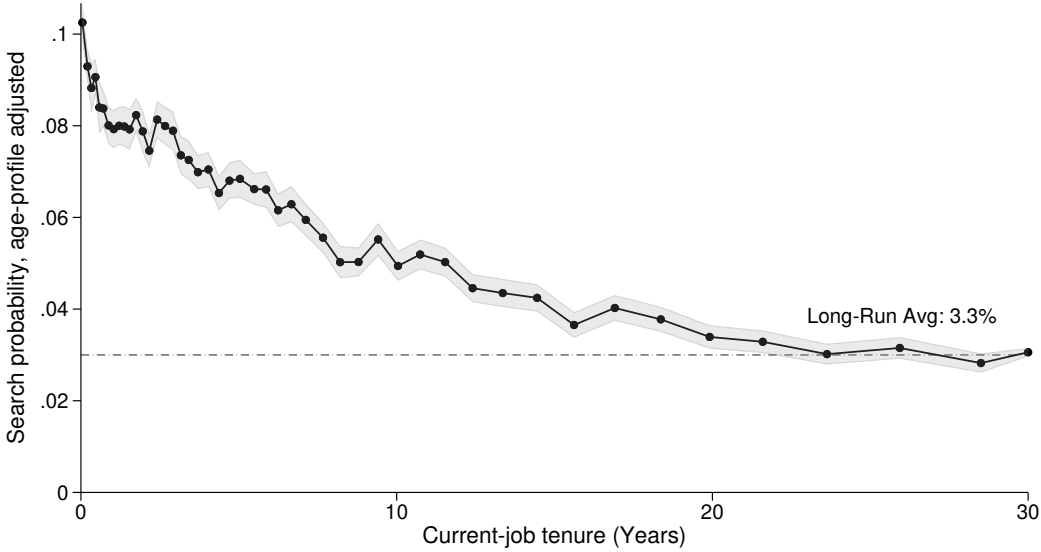
- Motives emphasize role of idiosyncratic preferences
- Overdispersion in search suggests persistent preference heterogeneity
  - ◇ Most workers **attached** → outside options permanently unattractive
- Leverage **employer tenure** to study heterogeneity
  - ◇ **Across matches**: Low levels of (voluntary) search persist in durable matches
  - ◇ **Within matches**: Search motivated by reset motive increases
  - → Non-pecuniary shocks to match value
- **Job satisfaction** around employer transitions
  - ◇ Transitions preceded by negative shocks
  - ◇ UK Data: Non-pecuniary, persistent returns to mobility
  - → Scope for **sorting into attachment** via search

⇒ **Search creates attachment via sorting on persistent, match-specific amenities**

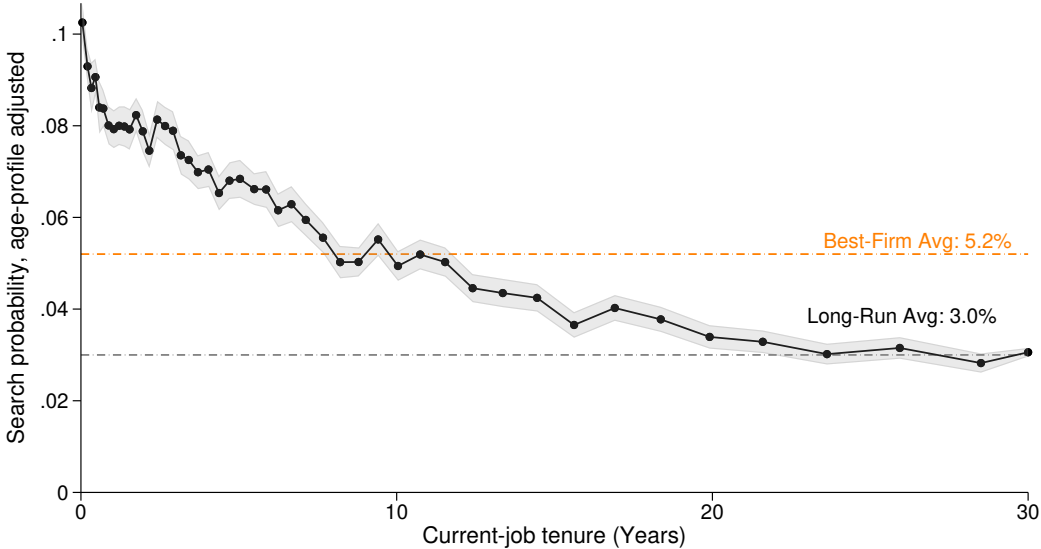
Cross-section: Search declines in tenure



Search stabilizes at around 3%



Low search compared to best-firm average



Within-Spell Motive Dynamics

Within-spell variation

- Search motives repeatedly observed
- Spell FEs absorb permanent match quality

Concern: Endogenous spell length

- Successful search ends employer spell
- Restrict to "**failed**" search at least 12 months before spell end

$$\widetilde{S}^{m}_{ist} = \beta_m \, \text{Tenure}_{ist} + \alpha_{is} + \varepsilon^{m}_{ist}$$

	Any Search
Cross-section	**-0.142**
	(0.008)

Within-Spell Motive Dynamics

Within-spell variation

- Search motives repeatedly observed
- Spell FEs absorb permanent match quality

Concern: Endogenous spell length

- Successful search ends employer spell
- Restrict to "**failed**" search at least 12 months before spell end

$$\widetilde{S}^{m}_{ist} = \beta_m \, \text{Tenure}_{ist} + \alpha_{is} + \varepsilon^{m}_{ist}$$

	Any Search	Try New
Cross-section	-0.142	**0.026**
	(0.008)	(0.005)

Within-Spell Motive Dynamics

Within-spell variation

- Search motives repeatedly observed
- Spell FEs absorb permanent match quality

Concern: Endogenous spell length

- Successful search ends employer spell
- Restrict to "**failed**" search at least 12 months before spell end

$$\widetilde{S}^{m}_{ist} = \beta_m \, \text{Tenure}_{ist} + \alpha_{is} + \varepsilon^{m}_{ist}$$

	Any Search	Try New	Other Motives
Cross-section	-0.142	0.026	**-0.168**
	(0.008)	(0.005)	(0.006)

Within-Spell Motive Dynamics

Within-spell variation

- Search motives repeatedly observed
- Spell FEs absorb permanent match quality

Concern: Endogenous spell length

- Successful search ends employer spell
- Restrict to "**failed**" search at least 12 months before spell end

$$\widetilde{S}^{m}_{ist} = \beta_m \, \text{Tenure}_{ist} + \alpha_{is} + \varepsilon^{m}_{ist}$$

	Any Search	Try New	Other Motives
Cross-section	-0.142	0.026	-0.168
	(0.008)	(0.005)	(0.006)
Within spell	**0.437**		
	(0.127)		

Within-Spell Motive Dynamics

Within-spell variation

- Search motives repeatedly observed
- Spell FEs absorb permanent match quality

Concern: Endogenous spell length

- Successful search ends employer spell
- Restrict to "**failed**" search at least 12 months before spell end

$$\widetilde{S}^{m}_{ist} = \beta_m \, \text{Tenure}_{ist} + \alpha_{is} + \varepsilon^{m}_{ist}$$

	Any Search	Try New	Other Motives
Cross-section	-0.142	0.026	-0.168
	(0.008)	(0.005)	(0.006)
Within spell	0.437	**0.347**	
	(0.127)	(0.087)	

Within-Spell Motive Dynamics

Within-spell variation

- Search motives repeatedly observed
- Spell FEs absorb permanent match quality

Concern: Endogenous spell length

- Successful search ends employer spell
- Restrict to "**failed**" search at least 12 months before spell end

$$\widetilde{S}^{m}_{ist} = \beta_m \, \text{Tenure}_{ist} + \alpha_{is} + \varepsilon^{m}_{ist}$$

	Any Search	Try New	Other Motives
Cross-section	-0.142	0.026	-0.168
	(0.008)	(0.005)	(0.006)
Within spell	0.437	0.347	**0.090**
	(0.127)	(0.087)	(0.099)

Within-Spell Motive Dynamics

Within-spell variation

- Search motives repeatedly observed
- Spell FEs absorb permanent match quality

Concern: Endogenous spell length

- Successful search ends employer spell
- Restrict to "**failed**" search at least 12 months before spell end

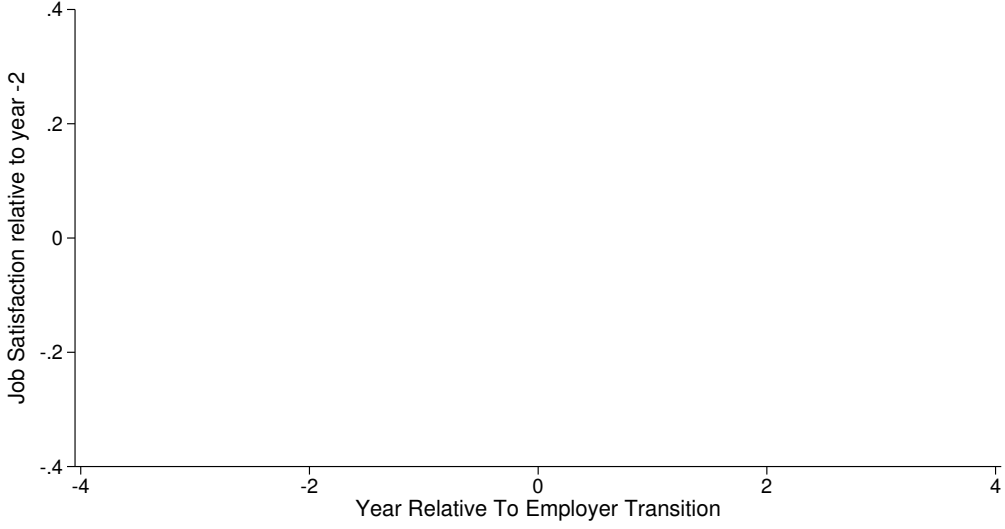
$$\widetilde{S}^{m}_{ist} = \beta_m \, \text{Tenure}_{ist} + \alpha_{is} + \varepsilon^{m}_{ist}$$

	Any Search	Try New	Other Motives
Cross-section	-0.142	0.026	-0.168
	(0.008)	(0.005)	(0.006)
Within spell	**0.437**	**0.347**	0.090
	(0.127)	(0.087)	(0.099)

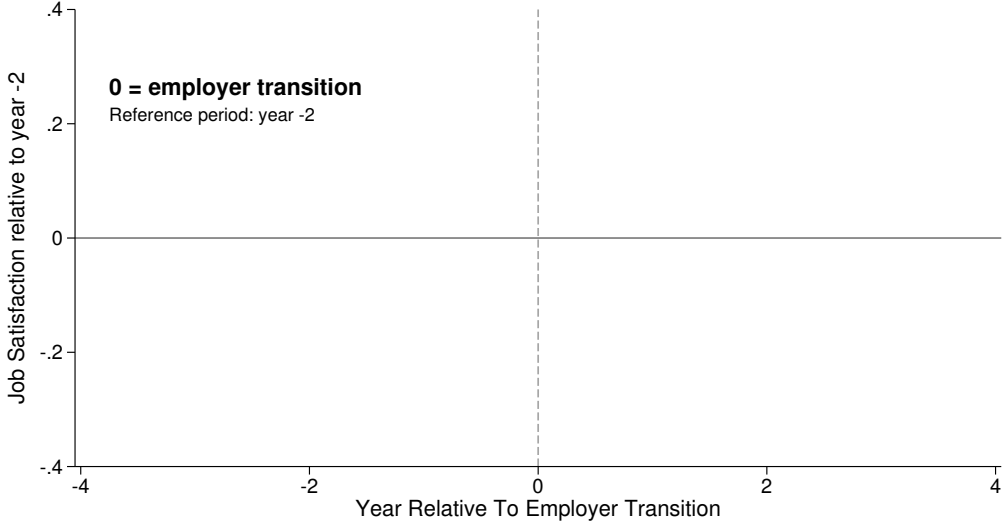
Within the same spell, the **tenure gradient reverses**, driven largely by **search to try something new**.

▸ Specification ▸ End Window ▸ Panel Length

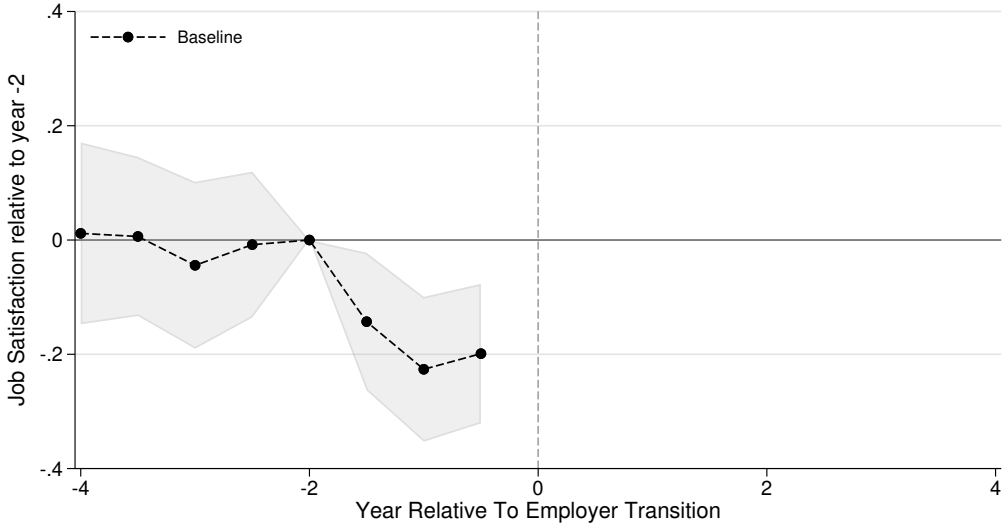
Job Satisfaction Around Employer Transitions



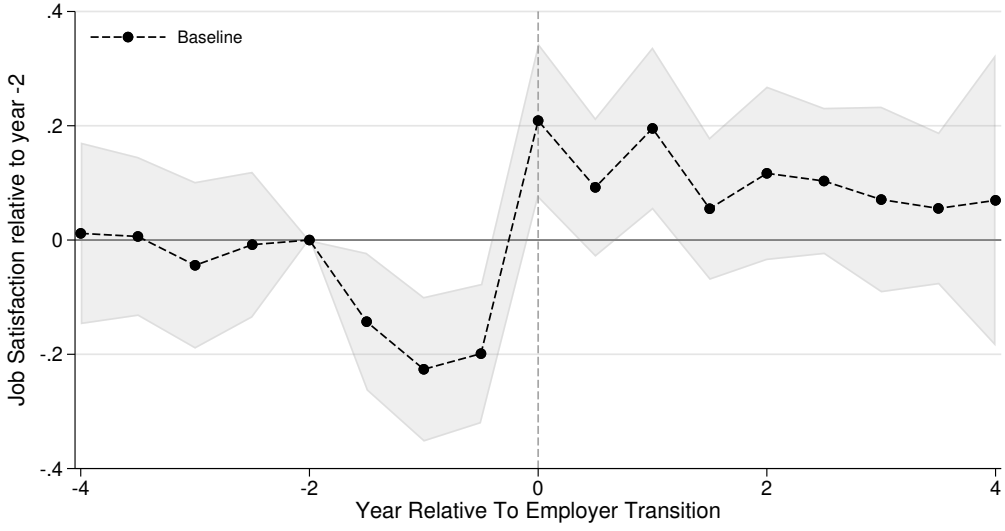
Align Workers Around Employer Transitions



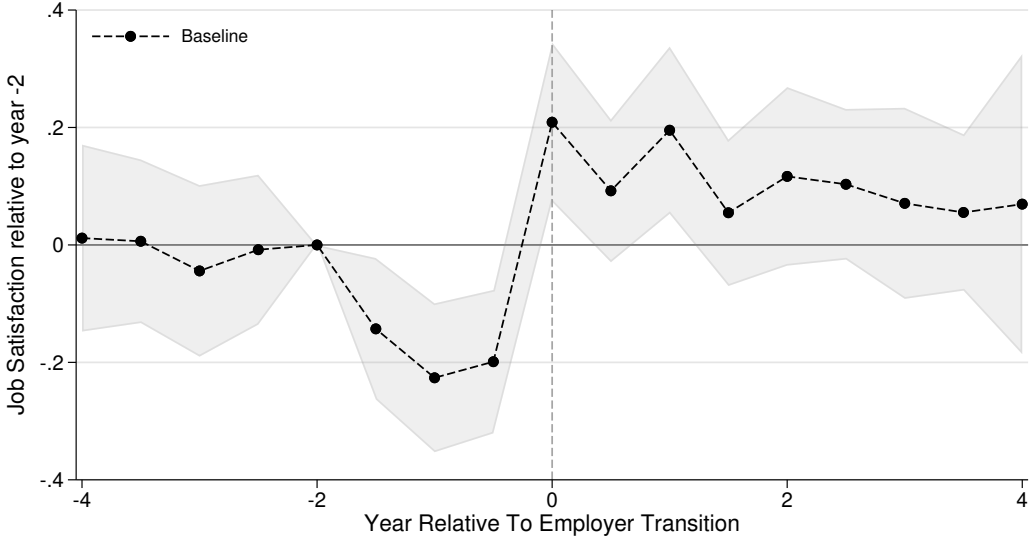
Job Satisfaction Declines Prior to Employer Transitions



Satisfaction Recovers After the Move



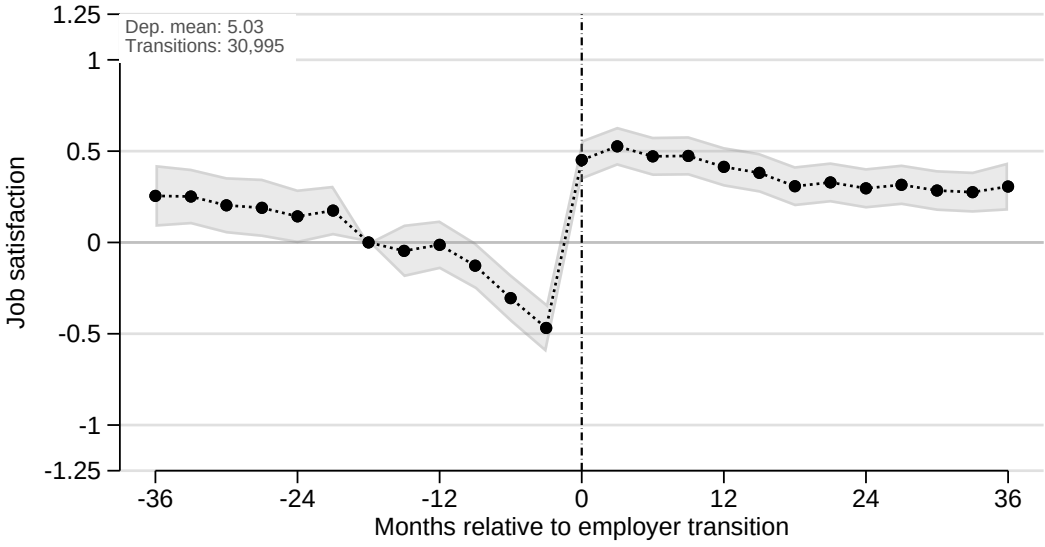
Concern: Short-Lived Origin Spells?



Pattern Persists for Long Origin Spells



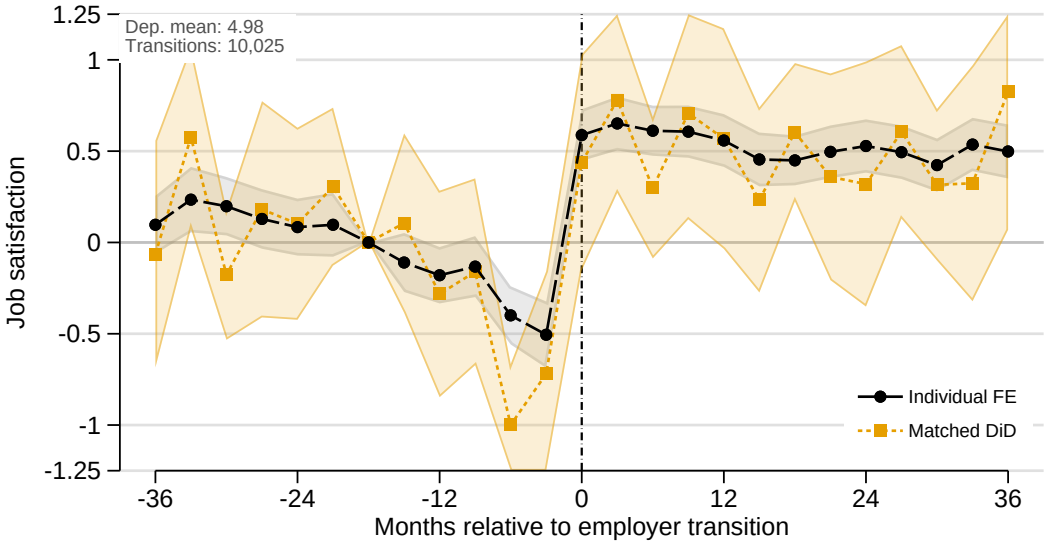
UK Data Exhibit the Same Pattern



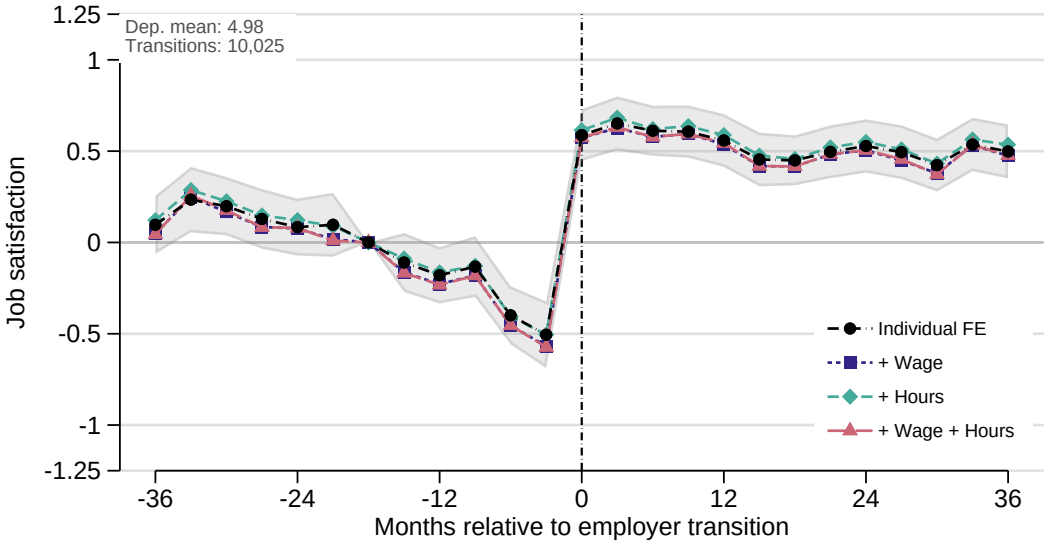
Pattern Persists Within Individuals



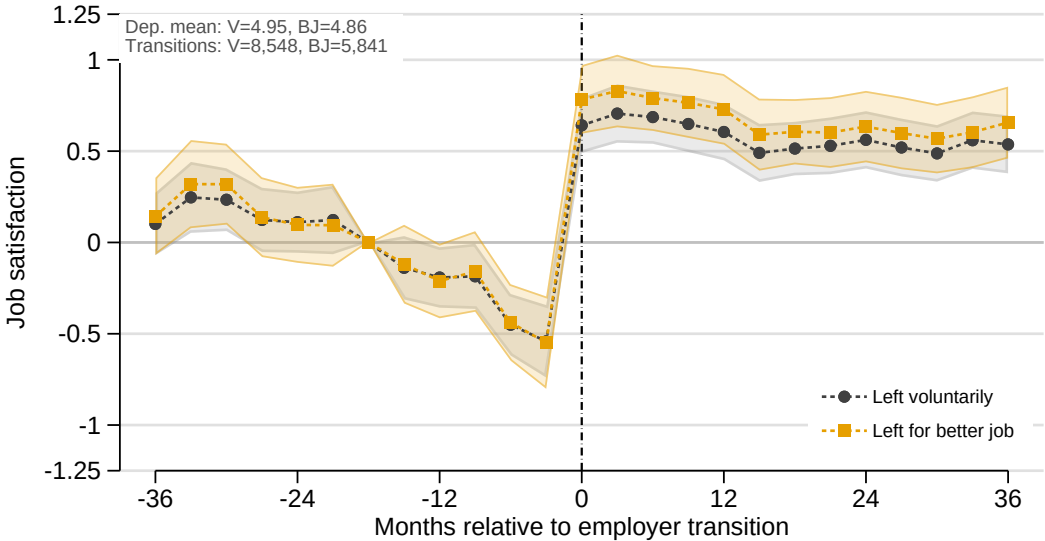
Balanced Panel + Match Workers on Pre-Move Satisfaction



Pattern Persists Controlling for Wages and Hours



Restrict to Voluntary Transitions



From the Empirics to the Model

- **Empirical findings: multidimensional returns and selection**
  - ◇ **Gradient:** Search declines with firm pay
  - ◇ **Motives:** Returns to search extend beyond pay
  - ◇ **Persistence:** Workers differ persistently in their willingness to leave
  - ◇ **Within-spell and event-study evidence:** Attachment evolves over time
- **To recover the response to a pay change, we need**
  1. **Search responsiveness to pay**, separated from non-pay attributes and match selection
  2. **Offer arrivals**, separated from willingness to accept
  3. **Match-value dynamics**, accounting for selective entry and retention
- **Dynamic discrete choice model of endogenous search**
  - ◇ Search, acceptance, and amenity changes generate the **equilibrium distribution of match quality** across firms
  - ◇ Match search gradients, mobility, hiring channels, and search dynamics to **jointly recover the parameters** (1)–(3)

Environment: Multidimensional Returns

- Time is discrete and measured in months.

$$\textbf{Worker State:} \qquad \mathbf{x} = (e, y, h, j, d, q)$$

  - e: education; y: firm type; h: hours mismatch; q: match amenity
  - (j, d): contract state and duration in that state

$$\textbf{Flow payoff:} \qquad u(\mathbf{x}) = \underbrace{a(e)}_{\text{Ability}} \times \underbrace{\psi(y)}_{\text{Firm pay}} \times \underbrace{(1-h)}_{\text{Hours}} + \underbrace{q}_{\text{Match amenity}}$$

- **Contracts capture employment protection**
  - Temporary, probationary, and permanent contracts, plus **advance notice**
  - Contracts evolve through renewal, conversion, entry into notice, and separation
  - **Anticipated job loss** encourages search; switching jobs entails **loss of security**

$\Rightarrow$ **Multiple reasons for search, rationalizes voluntary pay cuts**

Match Value Evolves Within the Job

- Each new match receives an idiosyncratic amenity draw

$$q \sim F^q, \qquad F^q : \text{Gumbel}, \quad \mathbb{E}[q] = 0, \quad \text{SD}(q) = \sigma_q$$

- Match amenity **resets** with monthly probability ζ
  - Reset draws are independent from F^q
  - $\zeta = 0$: permanent match amenity
  - $\zeta = 1$: independently redrawn every month

$\rightarrow$ **The value of an ongoing match can change over time**

Offer Acceptance Depends on Preference Heterogeneity

- Let $V^R(\mathbf{x})$ denote the **continuation value** after rejecting an outside offer
  Before contract transitions, separation risk, and amenity reset are realized
- A (ranked) offer is drawn from education-specific distribution $\hat{\mathbf{x}} \sim F(\cdot \mid e)$
  As before search does not change the offer distribution (!)
- **Workers accept** if $V(\hat{x}) \geqslant V^R(x)$
- A lower-ranked employer can offer a more valuable match

$$\mathcal{A}(\mathbf{x}) = \int \mathbb{1}\left[V(\hat{\mathbf{x}}) \geqslant V^R(\mathbf{x})\right] F_q(q)dq \tag{0.1}$$

$\Rightarrow$ **Preference heterogeneity shapes how offer acceptance responds to firm pay.**

Who Generates Outside Offers?

- Workers choose **active search** $s \in \{0, 1\}$

$$\lambda(s) = \lambda_0 + \lambda_1 s$$

- **Search cost:** Deterministic cost κ plus transitory shocks ϵ
  - $\epsilon \sim \text{Logistic}(0, \sigma_s)$, observed before the search decision
  - I.i.d draw each month

- Let $\Omega(\mathbf{x})$ denote the **expected gain** conditional on receiving an offer

$$\textbf{Search propensity:} \qquad \Pr(s = 1 \mid \mathbf{x}) = \Lambda\left(\frac{\widetilde{\beta}\lambda_1\Omega(\mathbf{x}) - \kappa}{\sigma_s}\right)$$

  $\rightarrow$ Dispersion of transitory costs; governs search responsiveness to net gains

$\Rightarrow$ **Workers generating offers select on incumbent match value**

Identification, Estimation, and Validation

- **Analytical identification results**
  - Mobility, search gradient, and search dynamics jointly identify key parameters
- **Joint estimation accounts for time aggregation**
  - Monthly decisions, quarterly interviews: search can change between observations
  - Construct model moments using the **survey's sampling scheme**
  - Match targeted moments by **weighted GMM**
- **Untargeted moments:** assess implications of the mechanism
- **External benchmarks**
  - Offers and acceptance: Faberman et al. (2022), ECB CES (Finland)
  - Incumbency premium: Caldwell et al. (2025)
  - Search response to job-finding rate

Identification: Search Technology and Preferences

- **Offer generation $\lambda(s)$ versus willingness to move $\mathcal{A}(s)$**
  - ◇ **Higher turnover among searchers** (+10pp): more offers or higher acceptance?
  - ◇ **Share of worker-initiated hires** provides discipline
- **Persistent match value σ_q versus temporary search costs σ_s**
  - ◇ **Repeated search** reveals **persistent willingness to switch jobs**
  - ◇ **Search in long-tenure jobs** informs **frequency of resets (ζ)**
- **Reduced-form elasticity of search with respect to firm pay**
  - ◇ A **shallow search gradient**: search-cost noise or preference heterogeneity?
  - ◇ Informs level of heterogeneity σ_q and search cost scale σ_s
- **Search cost κ**
  - ◇ **Aggregate search level** pins down **deterministic search costs κ**

▸ Offer arrivals ▸ Persistence ▸ Match heterogeneity

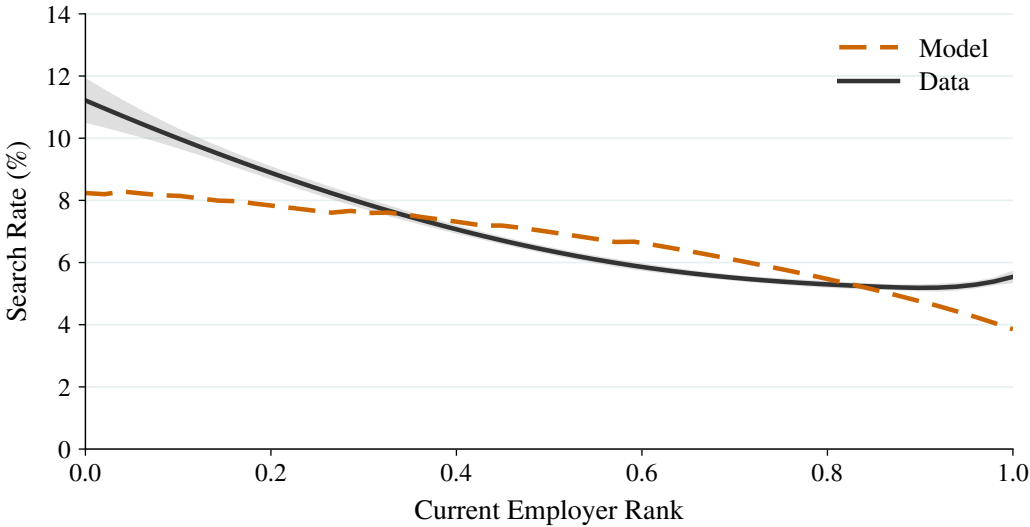
Parameter Estimates

Parameter	Description	Value
κ	Search cost	1.526
σ_s	Search-choice shock scale	0.232
σ_q	Match-amenity dispersion	0.614
ζ	Amenity reset probability	0.0062
(λ_0, λ_1)	Contact rates, employed	(0.023, 0.108)
$(\lambda_0^u, \lambda_1^u)$	Contact rates, unemployed	(0.037, 0.124)

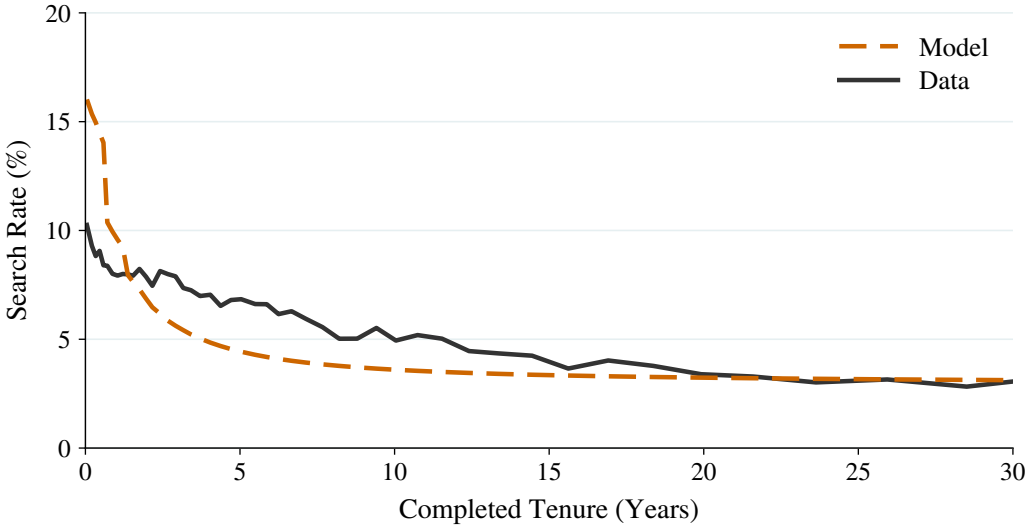
- **Search costs are high**
  - Average ex post search cost is $\sim$ 45 percent of monthly earnings
- Match amenities are **highly persistent** – Expected reset every 13.4 years

$\rightarrow$ **Substantial scope to sort on amenities, but sorting is costly**

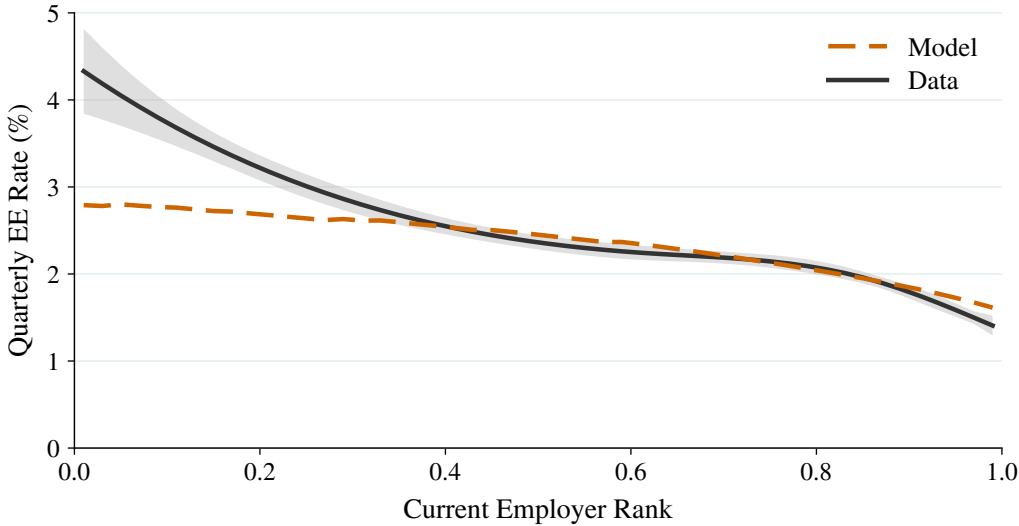
Model Fit: Decline in Search



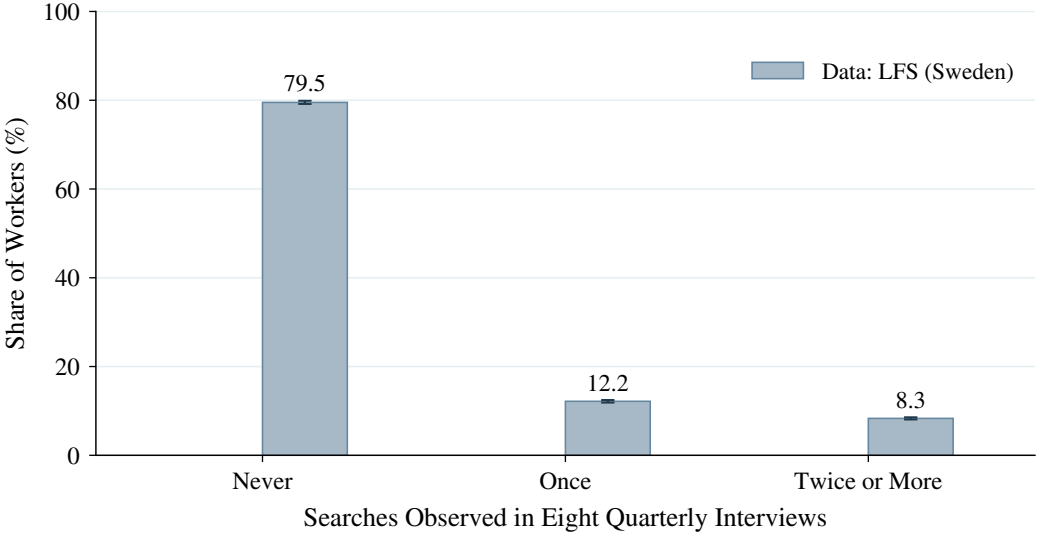
Model Fit: Search and Tenure



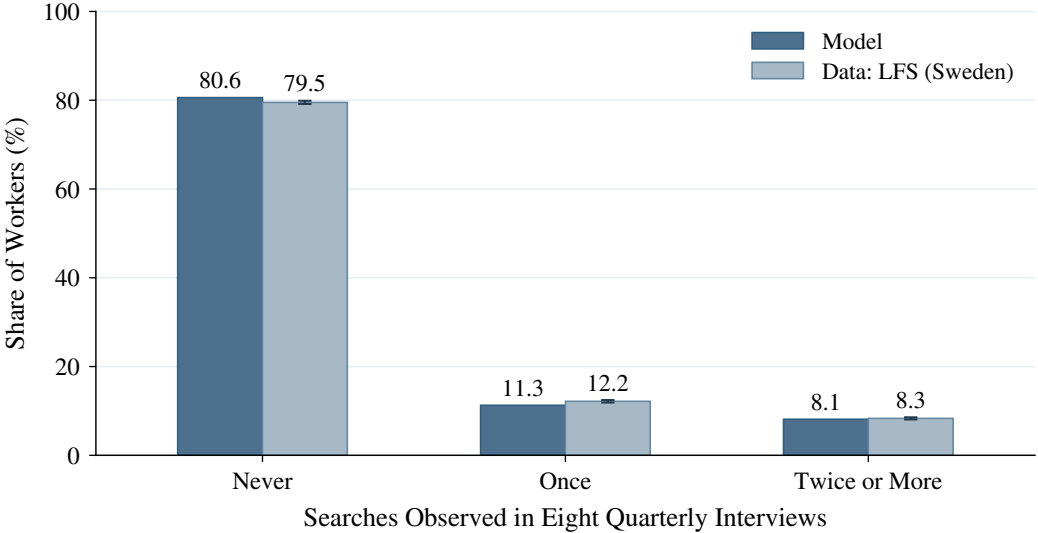
Model Fit: Employer-Employer Mobility



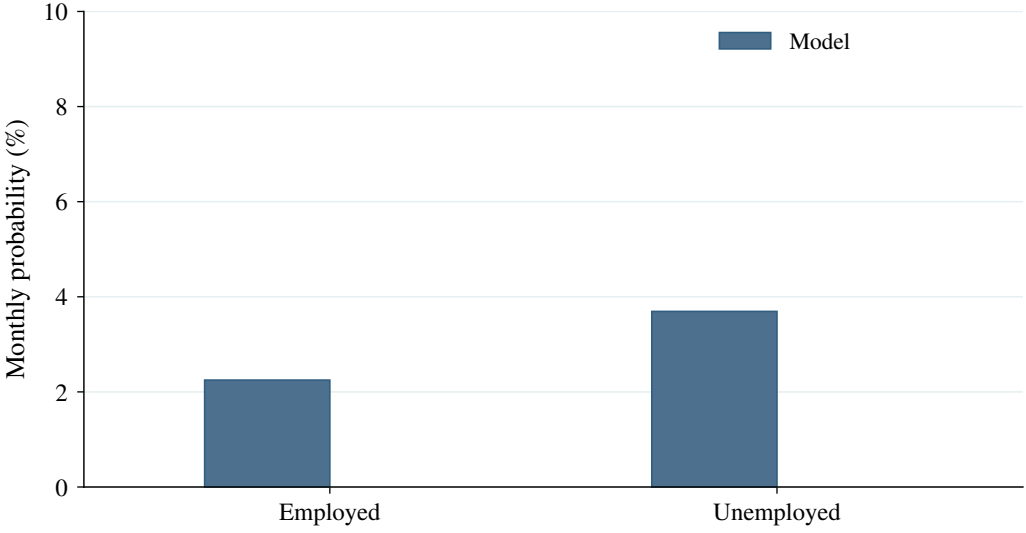
Validation: Model Matches Long-Run Overdispersion



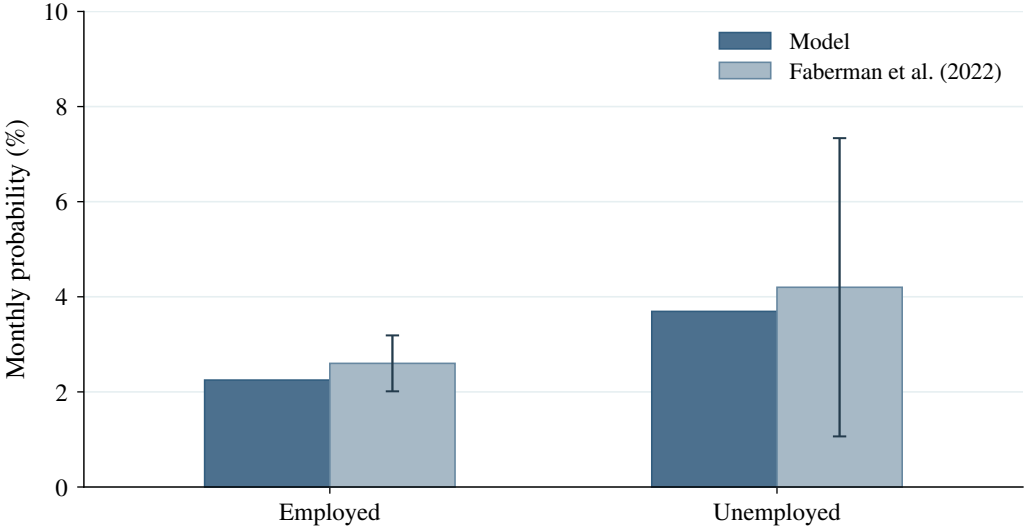
Validation: Model Matches Long-Run Overdispersion



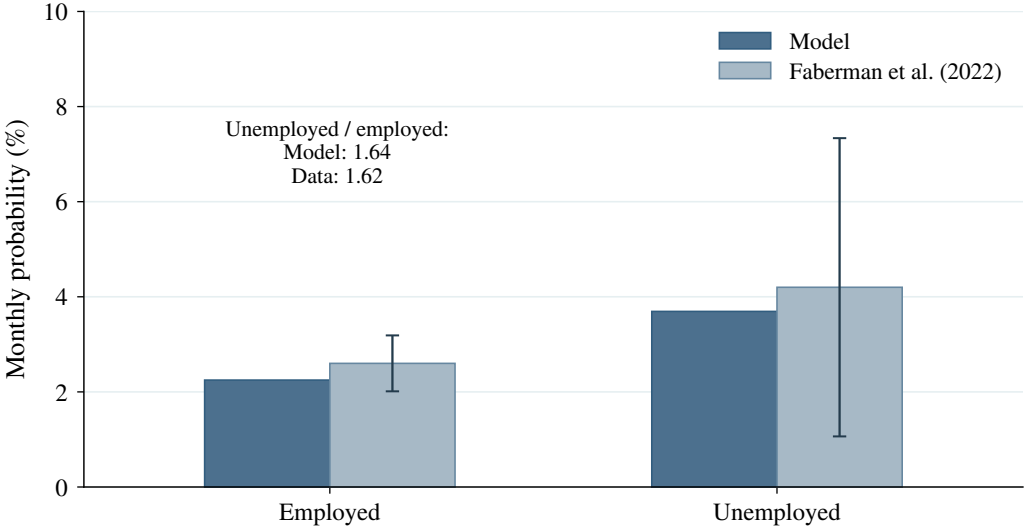
Validation: Unsolicited Offer Arrival



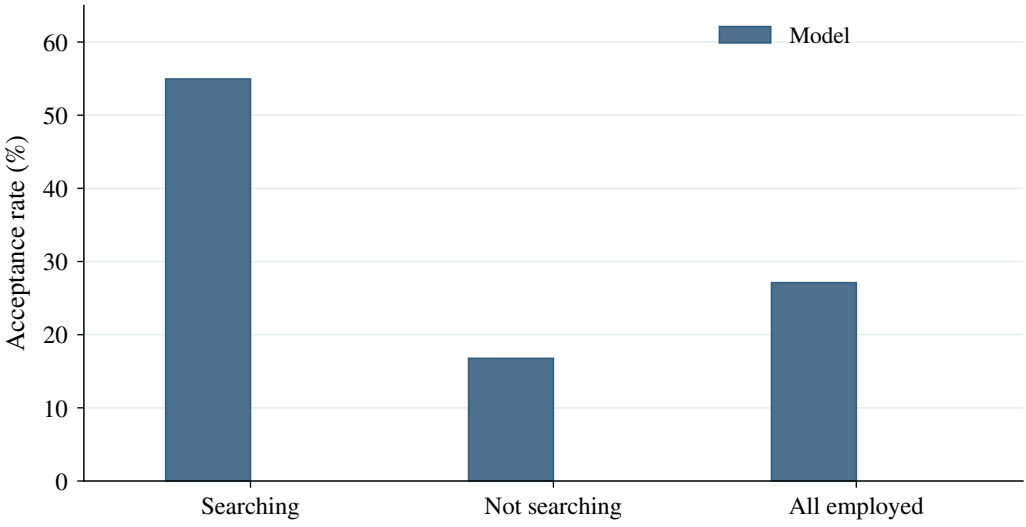
Validation: Unsolicited Offer Arrival



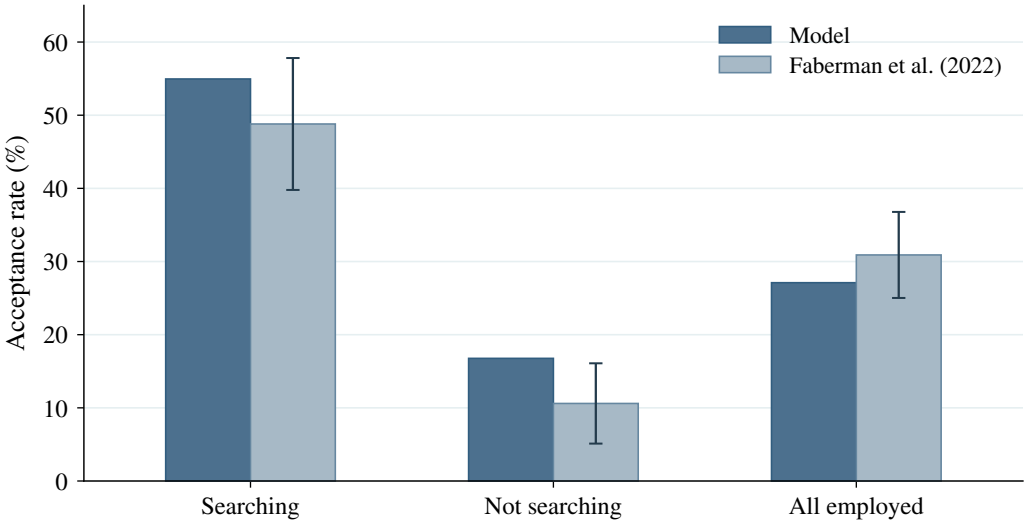
Validation: Unsolicited Offer Arrival



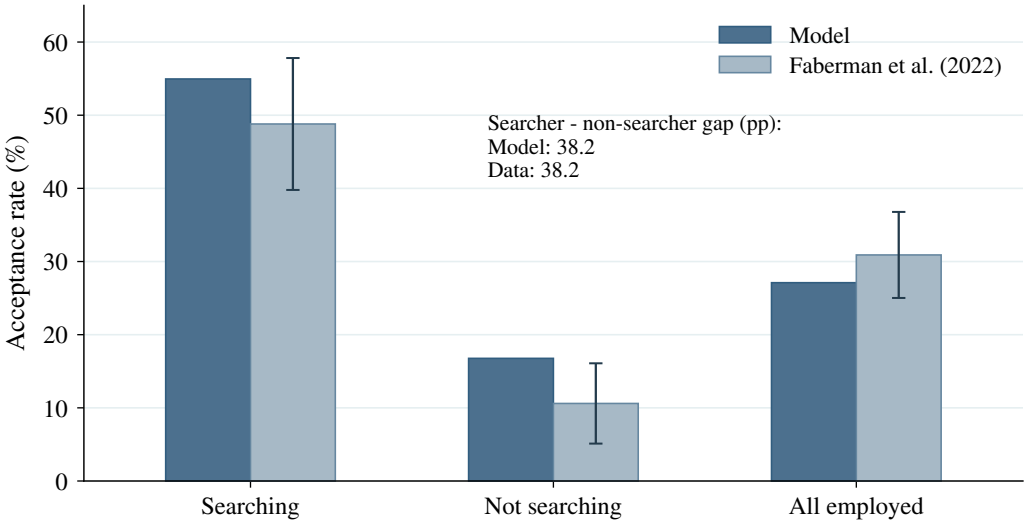
Validation: Conditional Acceptance Rates Align



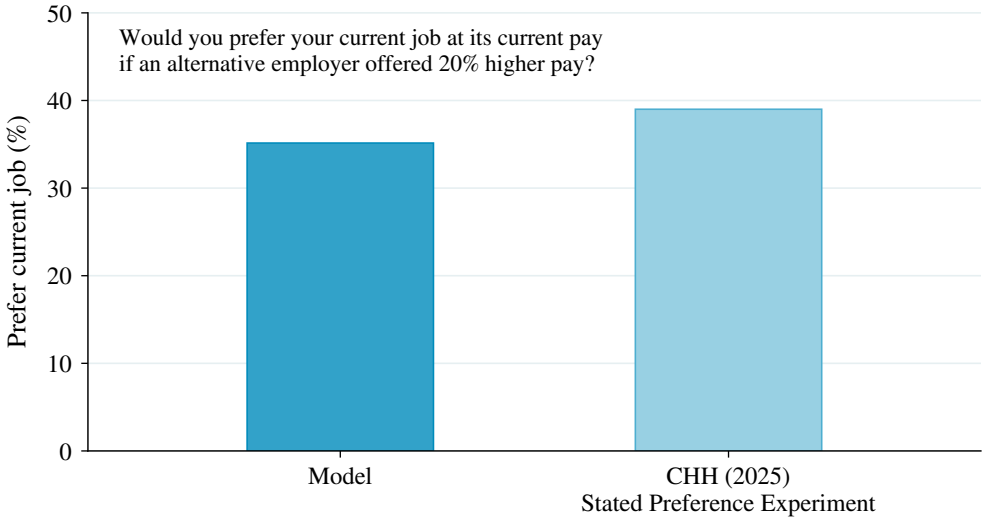
Validation: Conditional Acceptance Rates Align



Validation: Conditional Acceptance Rates Align



Validation: Staying Despite Higher Outside Pay



Validation: Response to Matching Efficiency

- **Model**: matching efficiency ϕ_t shifts all contact rates

$$\lambda_t(s) = \phi_t \left(\lambda_0 + \lambda_1 s\right)$$

  - ◇ Unemployed accept almost all offers
  - ⇒ Cyclical variation in unemployed job-finding rate measures ϕ_t
  - ◇ Higher ϕ_t raises return to search → size of response governed by σ_s

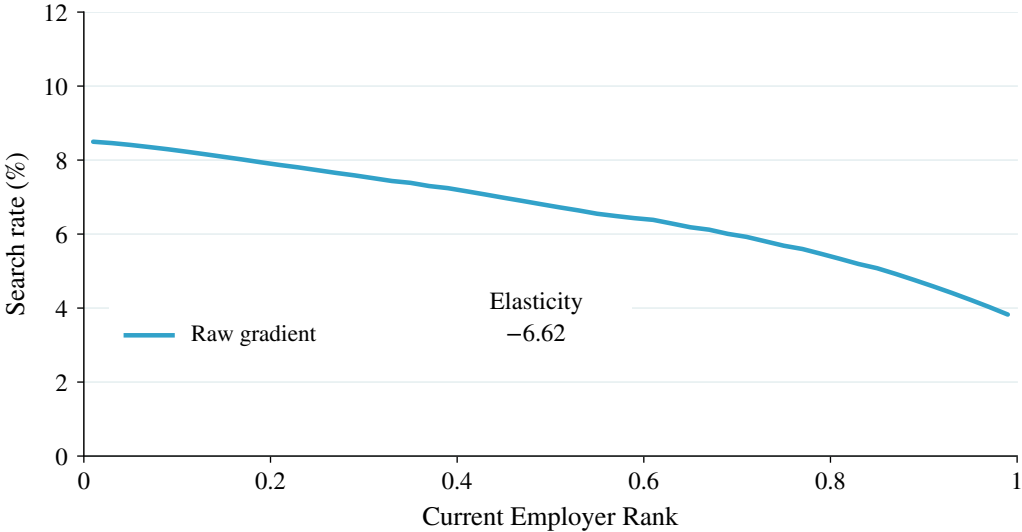
- **Data**: Regress **voluntary search** on the **cyclical job-finding rate**

$$s_{it}^{vol} = \beta\, \hat{f}_t^{U} + \alpha_{spell} + \delta_{m(t)} + \varepsilon_{it}$$

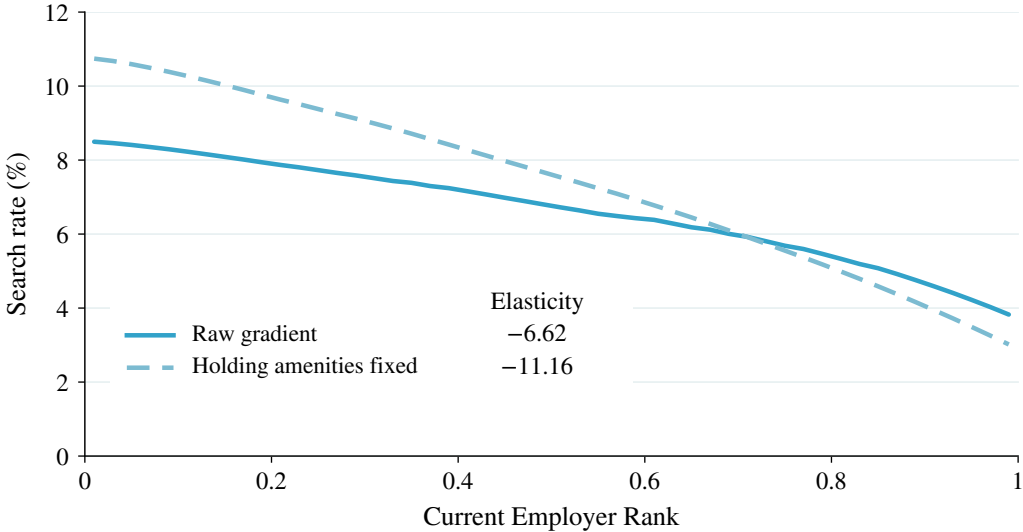
  - ◇ **Spell Fixed Effects**: Hold match quality fixed as the pool of employed changes
  - ◇ Voluntary motives: Exclude precautionary search driven by layoff risk

- **Untargeted**: **Data**: $\beta = \mathbf{0.366}$, CI : [0.232, 0.500] **Model**: $\beta = \mathbf{0.367}$

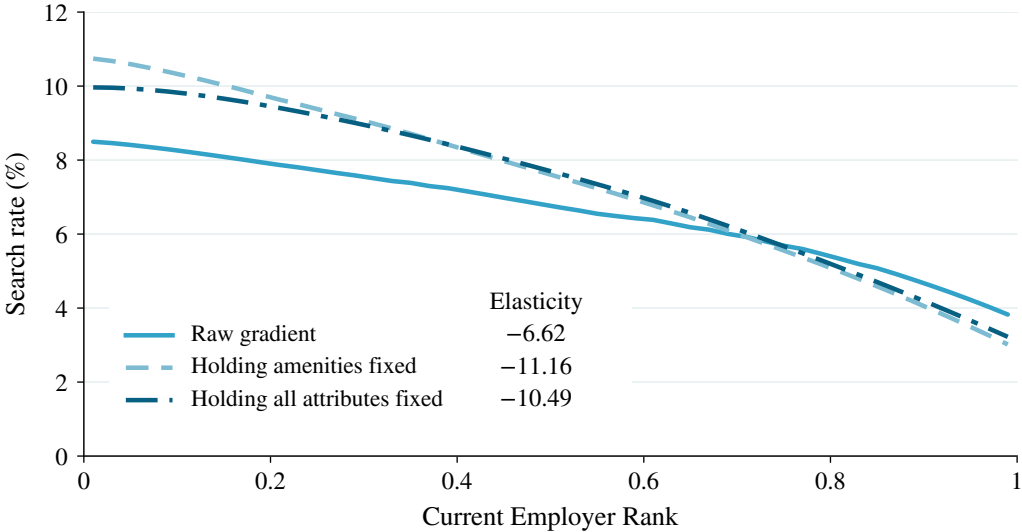
Selection and the Search–Pay Gradient



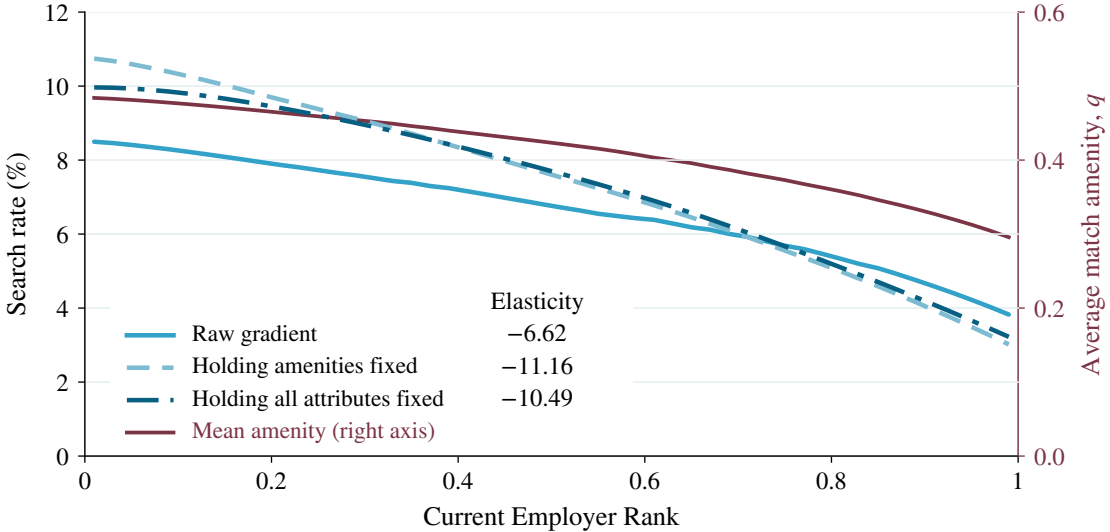
Selection and the Search–Pay Gradient



Selection and the Search–Pay Gradient



Selection and the Search–Pay Gradient



Search Strengthens the Retention Response

	Baseline search rule frozen	Endogenous search
Recruitment, ε_R	2.08	2.08
Retention contribution, $-\varepsilon_{sep}$	1.62	**2.95**
Employment, ε_N	3.71	**5.03**
Implied Markdown, $1/\varepsilon_N$ (%)	27.0	**19.9**

- Higher pay also reduces incumbents' search, strengthening the retention response

→ **Endogenous search creates an additional retention–recruitment wedge**

Decomposing the Amplification

Quit channel

$$\mathcal{C}_{\text{quit}} = \underbrace{\theta^{Q}}_{\substack{\text{Active hire share} \\ 0.454}} \left[\underbrace{-\varepsilon^{\bar{s}}}_{\substack{\text{Search elasticity} \\ 10.05}} + \underbrace{\frac{\mathrm{Cov}_s(A, -\varepsilon^s)}{\mathbb{E}_s[A]}}_{\substack{\text{Search–acceptance covariance} \\ -1.39}} \right] = \mathbf{3.93}.$$

- **Covariance:** Workers more willing to accept offers reduce search proportionally less.

Decomposing the Amplification

Quit channel

$$\mathcal{C}_{\text{quit}} = \underbrace{\theta^{Q}}_{\substack{\text{Active hire share}\\ 0.454}} \left[\underbrace{-\varepsilon^{\bar{s}}}_{\substack{\text{Search elasticity}\\ 10.05}} + \underbrace{\frac{\mathrm{Cov}_s(A, -\varepsilon^s)}{\mathbb{E}_s[A]}}_{\substack{\text{Search–acceptance covariance}\\ -1.39}} \right] = \mathbf{3.93}.$$

- **Covariance:** Workers more willing to accept offers reduce search proportionally less.

Stationary employment

$$\Delta\varepsilon^{N} = \underbrace{\bar{Q}\bar{S}^{-1}}_{\substack{\text{Quit share}\\ 0.515}} \mathcal{C}_{\text{quit}} + \underbrace{\mathcal{C}_{\text{workforce}}}_{\substack{\text{Workforce adjustment}\\ -0.687}} + \underbrace{\mathcal{R}}_{\substack{\text{Residual}\\ -0.016}} = \mathbf{1.32}.$$

Decomposing the Amplification

Quit channel

$$\mathcal{C}_{\text{quit}} = \underbrace{\theta^{Q}}_{\substack{\text{Active hire share}\\ 0.454}} \left[\underbrace{-\varepsilon^{\bar{s}}}_{\substack{\text{Search elasticity}\\ 10.05}} + \underbrace{\frac{\text{Cov}_s(A, -\varepsilon^{s})}{\mathbb{E}_s[A]}}_{\substack{\text{Search–acceptance covariance}\\ -1.39}} \right] = \mathbf{3.93}.$$

- **Covariance:** Workers more willing to accept offers reduce search proportionally less.

Stationary employment

$$\Delta\varepsilon^{N} = \underbrace{\bar{Q}\bar{S}^{-1}}_{\substack{\text{Quit share}\\ 0.515}} \mathcal{C}_{\text{quit}} + \underbrace{\mathcal{C}_{\text{workforce}}}_{\substack{\text{Workforce adjustment}\\ -0.687}} + \underbrace{\mathcal{R}}_{\substack{\text{Residual}\\ -0.016}} = \mathbf{1.32}.$$

- **Workforce adjustment:** Higher pay retains less-attached workers, whose matches remain fragile.

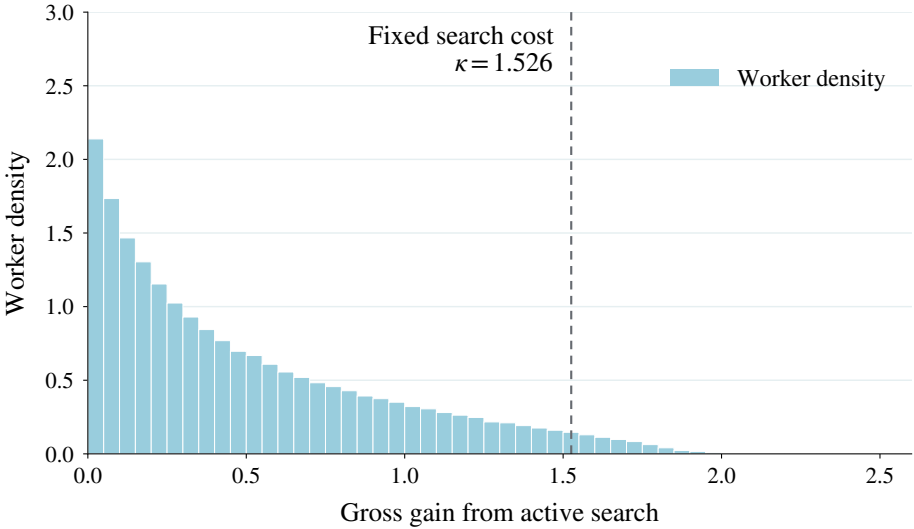
Fixed Costs and Workforce Adjustment

- Model estimates a large fixed cost to search
- Hence, small changes in pay only shift **marginal workers**
- Marginal workers have **low retention probabilities** (even after search reduction)
- Match quality is persistent $\rightarrow$ marginality carries into future

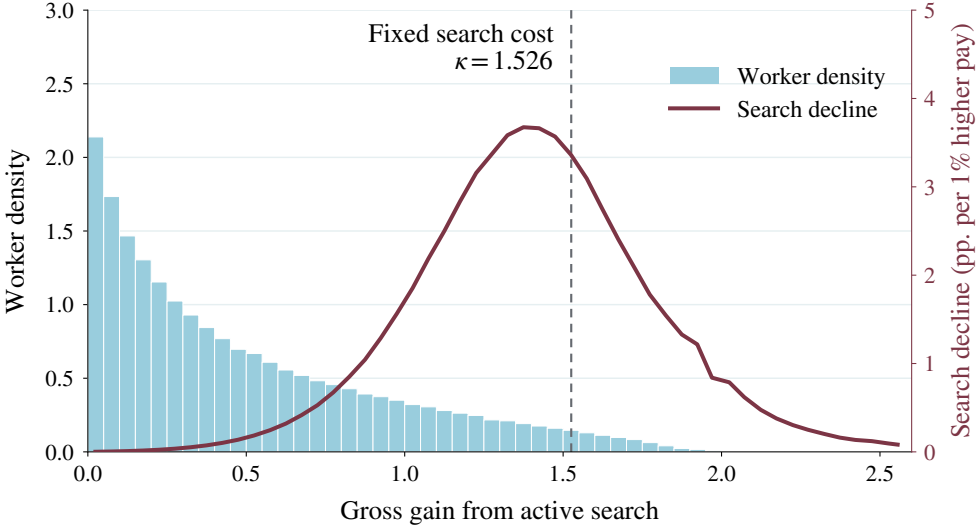
$\Rightarrow$ **Pay rise disproportionally retains high-turnover workers**

- Importance of accounting for **heterogeneity and dynamics**

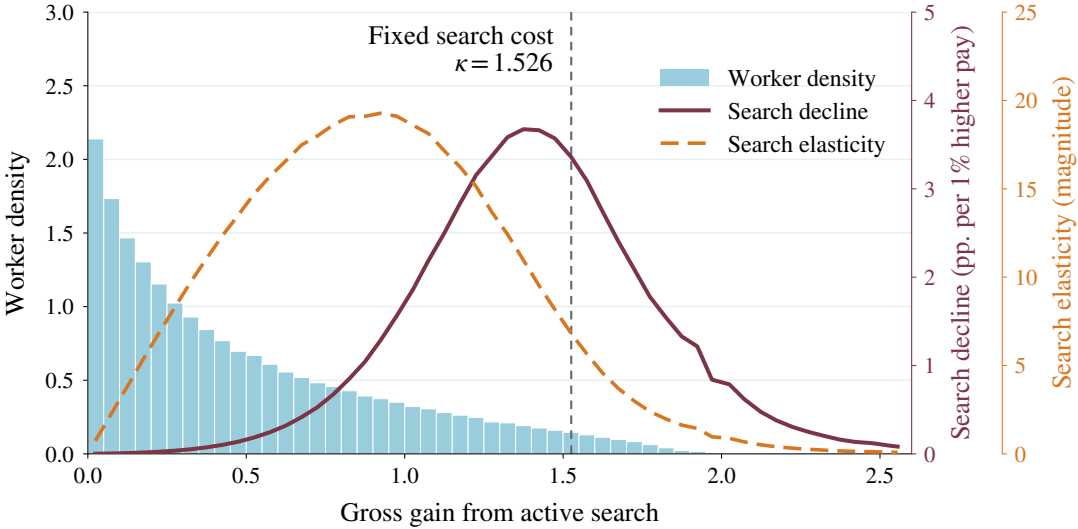
Heterogeneous Search Response



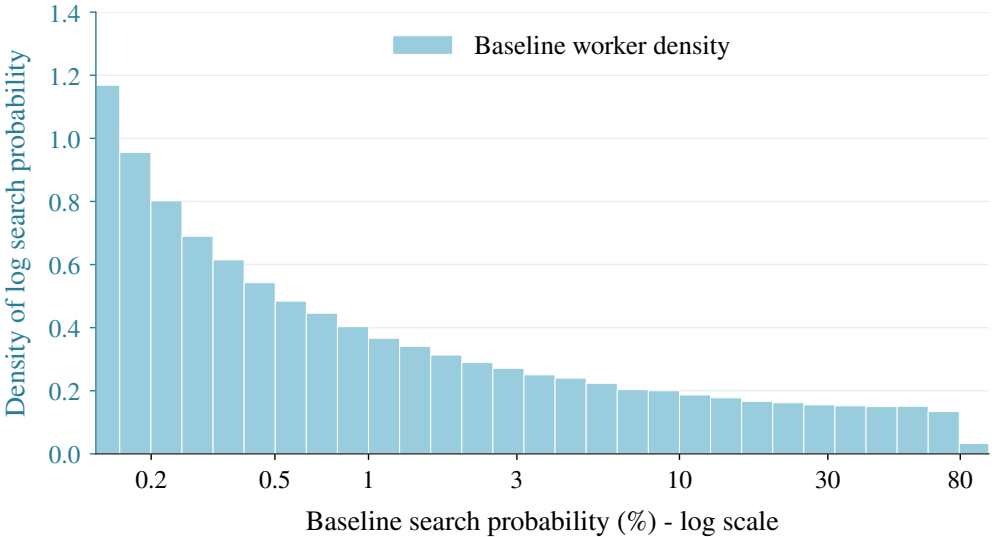
Heterogeneous Search Response



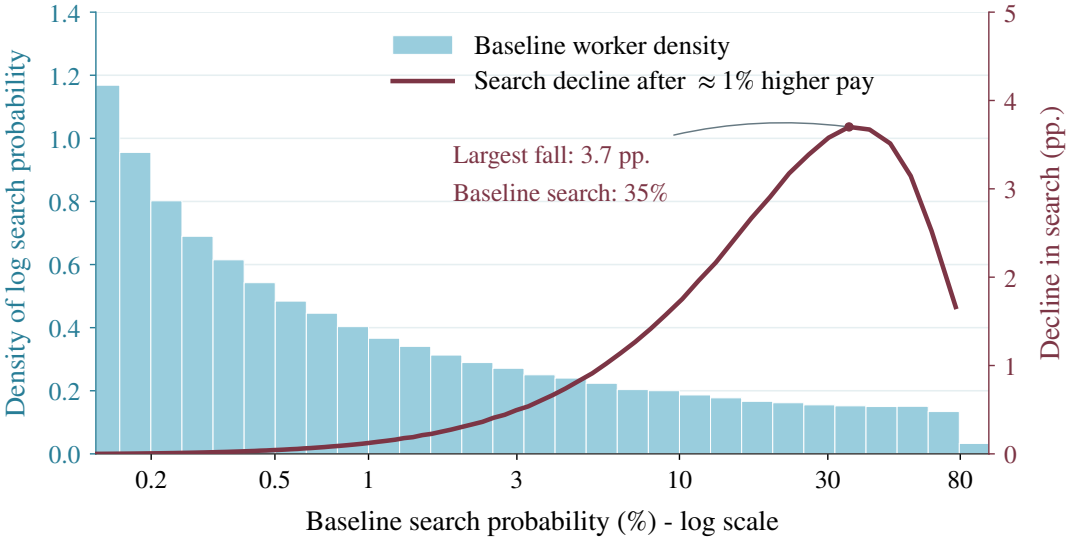
Heterogeneous Search Response



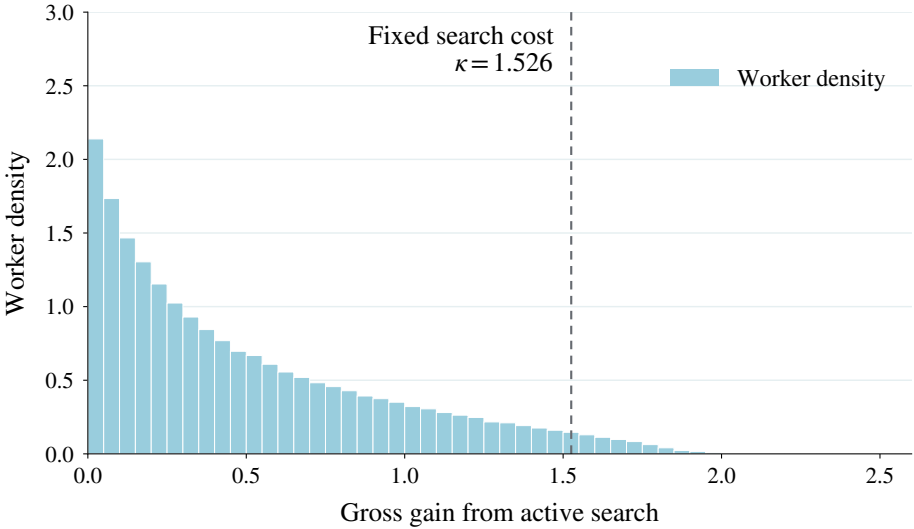
Search Responses to Higher Firm Pay



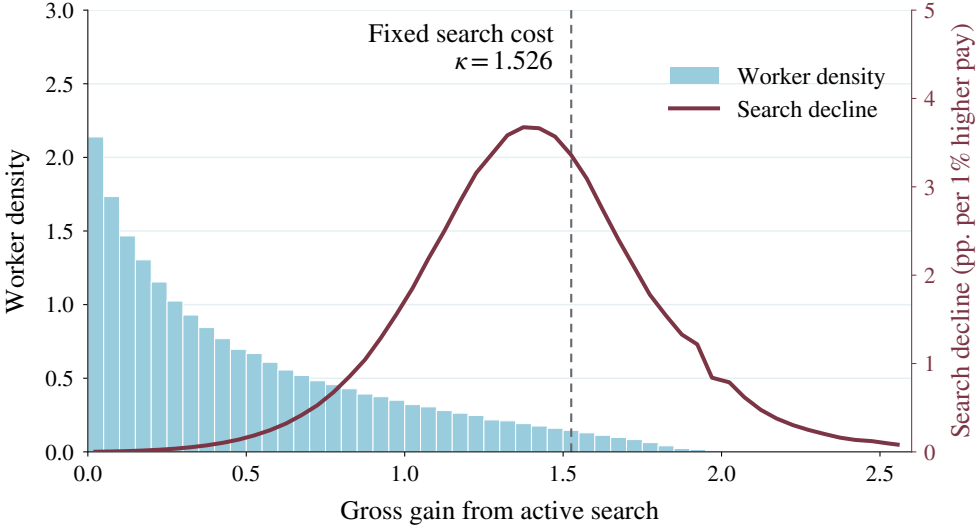
Search Responses to Higher Firm Pay



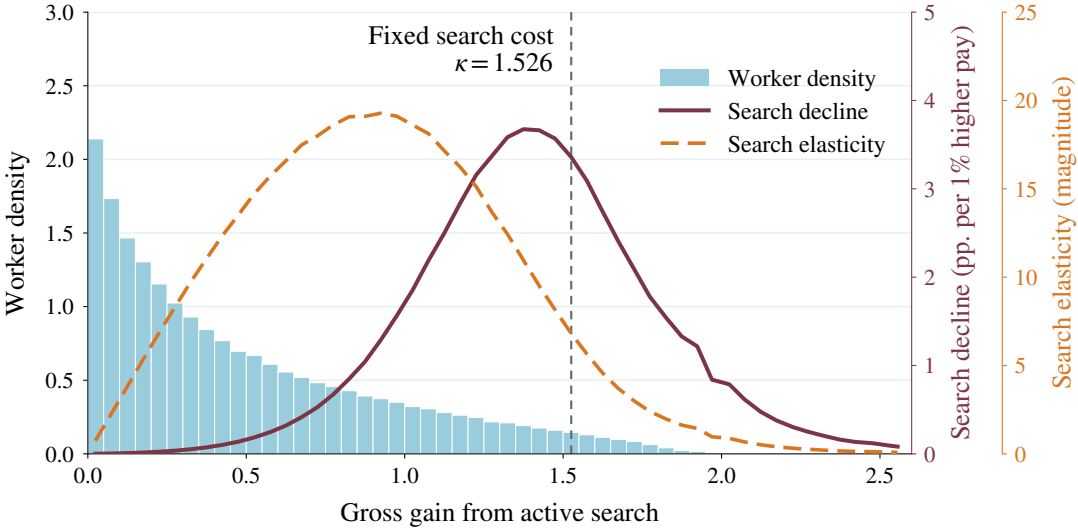
Heterogeneous Search Response



Heterogeneous Search Response



Heterogeneous Search Response



Search Responses to Higher Firm Pay

Panel B. Decomposition of the increase in ε_w^N

Object	Component	Value	Contribution
$\mathbb{E}_p[D]$	Mean search response	10.197	+2.357
$\mathrm{Cov}_p(A, D)/\mathbb{E}_p[A]$	Search–acceptance covariance	−1.381	−0.319
$\mathbb{E}_{pA}\left[D(x)\left\{\bar{S}(1-\delta(x))(\Pi T)(x)-1\right\}\right]$	Workforce adjustment	−3.097	−0.716
$\mathcal{R}_{\text{other}}$	Remaining corrections		+0.006
$\Delta\varepsilon_w^R$	Recruitment		+0.004
$\boldsymbol{\Delta\varepsilon_w^N}$	**Total**		**+1.332**

Conclusion

- **Search effort responds to workers' current jobs**
  - ◇ Search decreases with employer quality
  - ◇ Residual search reflects idiosyncratic, non-pecuniary match value
  - ◇ Match-specific job values are persistent, but evolve over time
- **Outside opportunities are endogenous**
  - ◇ Search determines access to outside offers
  - ◇ Workers selecting into search differ in their willingness to move
- **Implications for firm labor supply**
  - ◇ Endogenous search strengthens the retention response to pay
  - ◇ Understanding heterogeneity of outside options key

$\Rightarrow$ **Firm labor supply depends the endogenous formation of workers' choice sets**

References I

Abowd, J. M., F. Kramarz, and D. N. Margolis (1999): "High Wage Workers and High Wage Firms," *Econometrica*, 67, 251–333.

Afrouzi, H., A. Blanco, A. Drenik, and E. Hurst (2026): "A Theory of How Workers Keep Up with Inflation," *The Quarterly Journal of Economics*, 141, 945–1004.

Bagger, J. and R. Lentz (2019): "An Empirical Model of Wage Dispersion with Sorting," *The Review of Economic Studies*, 86, 153–190.

Belot, M., P. Kircher, and P. Muller (2022): "How Wage Announcements Affect Job Search—A Field Experiment," *American Economic Journal: Macroeconomics*, 14, 1–67.

Berger, D., K. Herkenhoff, A. R. Kostøl, and S. Mongey (2024): "An Anatomy of Monopsony: Search Frictions, Amenities, and Bargaining in Concentrated Markets," *NBER Macroeconomics Annual*, 38, 1–47.

Berger, D., K. Herkenhoff, and S. Mongey (2022): "Labor Market Power," *American Economic Review*, 112, 1147–1193.

References II

CALDWELL, S., I. HAEGELE, AND J. HEINING (2025): "Firm Pay and Worker Search," Tech. rep., National Bureau of Economic Research.

——— (2026): "Firm Pay, Amenities, and Inequality," Tech. rep., National Bureau of Economic Research.

CARD, D. (2022): "Who Set Your Wage?" *American Economic Review*, 112, 1075–1090.

CHRISTENSEN, B. J., R. LENTZ, D. T. MORTENSEN, G. R. NEUMANN, AND A. WERWATZ (2005): "On-the-Job Search and the Wage Distribution," *Journal of Labor Economics*, 23, 31–58.

DAL BÓ, E., F. FINAN, AND M. A. ROSSI (2013): "Strengthening State Capabilities: The Role of Financial Incentives in the Call to Public Service," *The Quarterly Journal of Economics*, 128, 1169–1218.

DATTA, N. (2023): "The Measure of Monopsony: The Labour Supply Elasticity to the Firm and Its Constituents," Discussion Paper 1930, Centre for Economic Performance, London School of Economics.

References III

EECKHOUT, J. AND I. LINDENLAUB (2019): "Unemployment Cycles," *American Economic Journal: Macroeconomics*, 11, 175–234.

FABERMAN, R. J., A. I. MUELLER, AND A. ŞAHIN (2026): "Job Search, Job Amenities and the Gender Pay Gap," Tech. rep., National Bureau of Economic Research.

FABERMAN, R. J., A. I. MUELLER, A. ŞAHIN, AND G. TOPA (2022): "Job Search Behavior among the Employed and Non-Employed," *Econometrica*, 90, 1743–1779.

FALCH, T. (2017): "Wages and Recruitment: Evidence from External Wage Changes," *ILR Review*, 70, 483–518.

FUJITA, S. (2010): *Reality of On-the-Job Search*, Federal Reserve Bank of Philadelphia.

HALL, R. E. AND A. I. MUELLER (2018): "Wage Dispersion and Search Behavior: The Importance of Nonwage Job Values," *Journal of Political Economy*, 126, 1594–1637.

HUMLUM, A., M. RASMUSSEN, AND E. K. ROSE (2025): "Firm Premia and Match Effects in Pay vs. Amenities," Tech. rep., National Bureau of Economic Research.

References IV

Jarosch, G. (2023): "Searching for Job Security and the Consequences of Job Loss," *Econometrica*, 91, 903–942.

Kaas, L., E. Lalé, and N. Siassi (2026): "Job Ladder and Wealth Dynamics in General Equilibrium," *Econometrica*, 94, 1449–1485.

Kline, P. (2025): "Labor Market Monopsony: Fundamentals and Frontiers," in *Handbook of Labor Economics*, Elsevier, vol. 6, chap. 8, 655–728.

Lamadon, T., M. Mogstad, and B. Setzler (2022): "Imperfect Competition, Compensating Differentials, and Rent Sharing in the US Labor Market," *American Economic Review*, 112, 169–212.

Lise, J. (2013): "On-the-Job Search and Precautionary Savings," *The Review of Economic Studies*, 80, 1086–1113.

Maestas, N., K. J. Mullen, D. Powell, T. Von Wachter, and J. B. Wenger (2023): "The Value of Working Conditions in the United States and the Implications for the Structure of Wages," *American Economic Review*, 113, 2007–2047.

References V

MANNING, A. (2021): "Monopsony in Labor Markets: A Review," *ILR Review*, 74, 3–26.

MAS, A. AND A. PALLAIS (2017): "Valuing Alternative Work Arrangements," *American Economic Review*, 107, 3722–3759.

MORCHIO, I. AND C. MOSER (2026): "The Gender Pay Gap: Micro Sources and Macro Consequences," *American Economic Review*, 116, 1765–1810.

NAGYPÁL, É. (2007): "Labor-Market Fluctuations and On-the-Job Search," *Available at SSRN 1088005*.

PILOSSOPH, L., J. M. RYNGAERT, AND J. J. WEDEWER (2026): "The Search Costs of Inflation in the Labor Market," Tech. rep., National Bureau of Economic Research.

POSTEL-VINAY, F. AND J.-M. ROBIN (2004): "To Match or Not to Match? Optimal Wage Policy with Endogenous Worker Search Intensity," *Review of Economic Dynamics*, 7, 297–330.

References VI

ROUSSILLE, N. AND B. SCUDERI (2026): "Bidding for Talent: A Test of Conduct in a High-Wage Labor Market," Working paper, January 2026.

SIMMONS, M. (2024): "Precautionary On-the-Job Search," *Available at SSRN 4685118*.

SORKIN, I. (2018): "Ranking Firms Using Revealed Preference," *The Quarterly Journal of Economics*, 133, 1331–1393.

TABER, C. AND R. VEJLIN (2020): "Estimation of a Roy/Search/Compensating Differential Model of the Labor Market," *Econometrica*, 88, 1031–1069.

WISWALL, M. AND B. ZAFAR (2018): "Preference for the Workplace, Investment in Human Capital, and Gender," *The Quarterly Journal of Economics*, 133, 457–507.

Identification: Offer Arrivals

Observed in the data:

- $f(s)$, $\bar{f}$: EE rates by search status and overall
- θ_e: Worker-initiated share of EE moves

Unobserved: Offer arrival rates (λ_0, λ_1) and acceptance rates by search status $A(s)$

$$\textbf{Offer arrivals:} \quad \lambda_0 = \frac{f(0)}{A(0)}, \qquad \lambda_1 = \frac{\theta_e \bar{f}}{\Pr(s=1)\, A(1)}$$

$$\textbf{Search premium:} \quad f(1) - f(0) = \underbrace{\frac{\theta_e \bar{f}}{\Pr(s=1)}}_{\text{Search-generated moves}} + \underbrace{f(0)\left[\frac{A(1)}{A(0)} - 1\right]}_{\text{Acceptance selection}}$$

$\rightarrow$ **Given acceptance rate anchor $A(0)$, moments discipline $(A(1),\ \lambda_0,\ \lambda_1)$**

Identification: Search Persistence

- **Benchmark:** other states fixed; $q \sim F_q$, redrawn with probability ζ.

$$\textbf{Persistent variance share:} \quad \omega_F \equiv \frac{\mathrm{Var}[\Pr(s = 1 \mid q)]}{\mathrm{Var}(s)}$$

- **Intuition:** Ceteris paribus, an increase in
  - ◇ Transitory search-cost variance ($\sigma_s \uparrow$) $\Rightarrow$ lower ω_F.
  - ◇ Persistent amenity dispersion ($\sigma_q \uparrow$) $\Rightarrow$ higher ω_F.

$$\textbf{Autocorrelation:} \quad \mathrm{Corr}(s_t, s_{t+k}) = \underbrace{(1-\zeta)^k}_{\text{Amenity persistence}} \times \underbrace{\omega_F}_{\text{Amenity share of variance}}, \qquad k \geqslant 1$$

$\rightarrow$ **Search persistence disciplines ω_F, conditional on ζ**

Identification: Match Heterogeneity

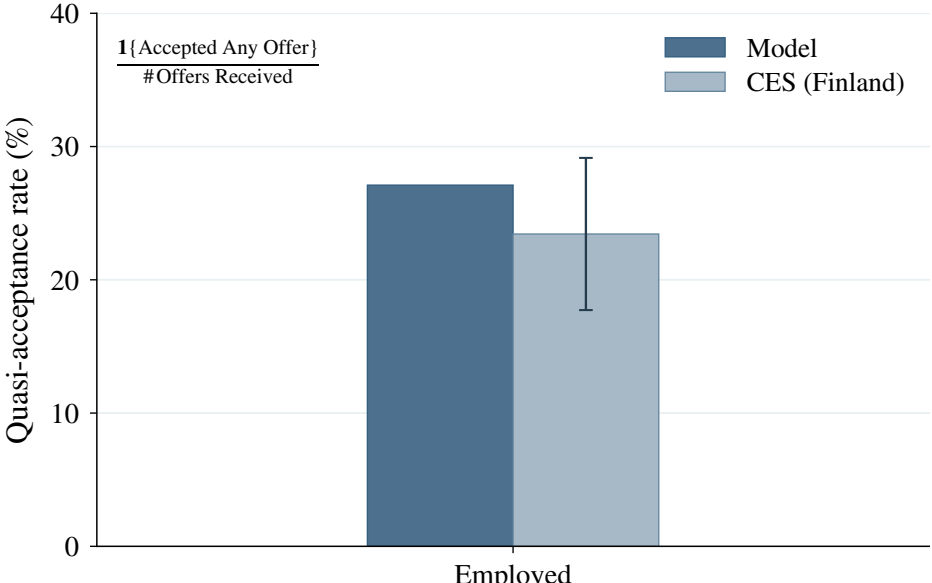
- $p(y) \equiv \Pr(s = 1 \mid y)$: search rate by firm type; $\ell(p) \equiv \log[p/(1-p)]$.

$$\frac{d\ell[p(y)]}{dy} = \underbrace{\frac{\bar{v}_y^M(y)}{\sigma_s}}_{\text{Behavioral response}} \times \underbrace{[1-\omega(y)]}_{\text{Heterogeneity attenuation}} + \underbrace{\frac{S(y)}{p(y)[1-p(y)]}}_{\text{Match selection}}$$

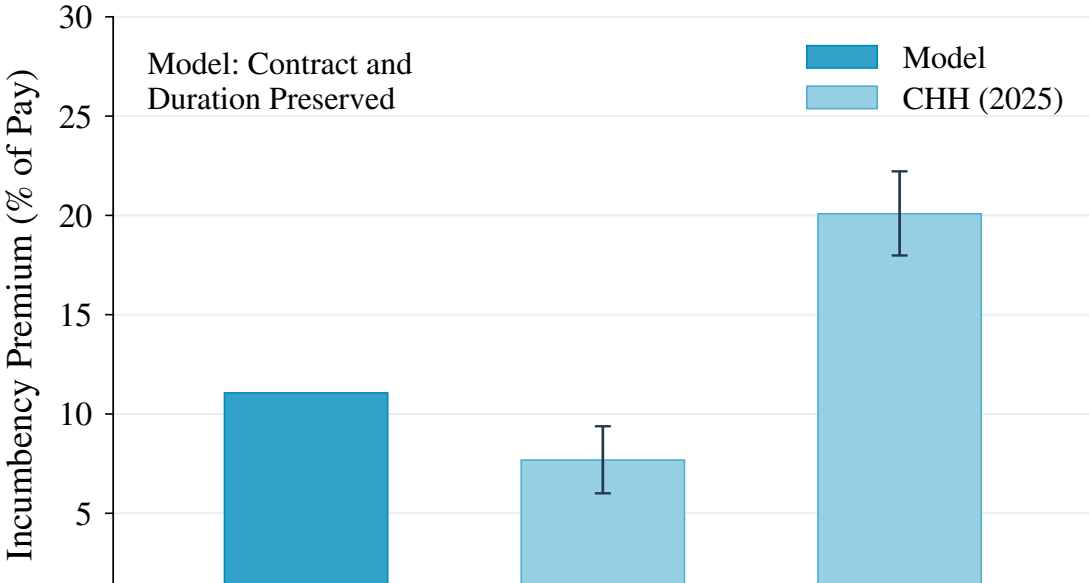
- $\bar{v}_y^M(y)$: Weighted average gradient of search gains.
- $\omega(y)$: Persistent variance share within firm type y.
- $S(y)$: Changes in match composition across firm types.

$\rightarrow$ **Persistence and gradients jointly discipline σ_q and σ_s**

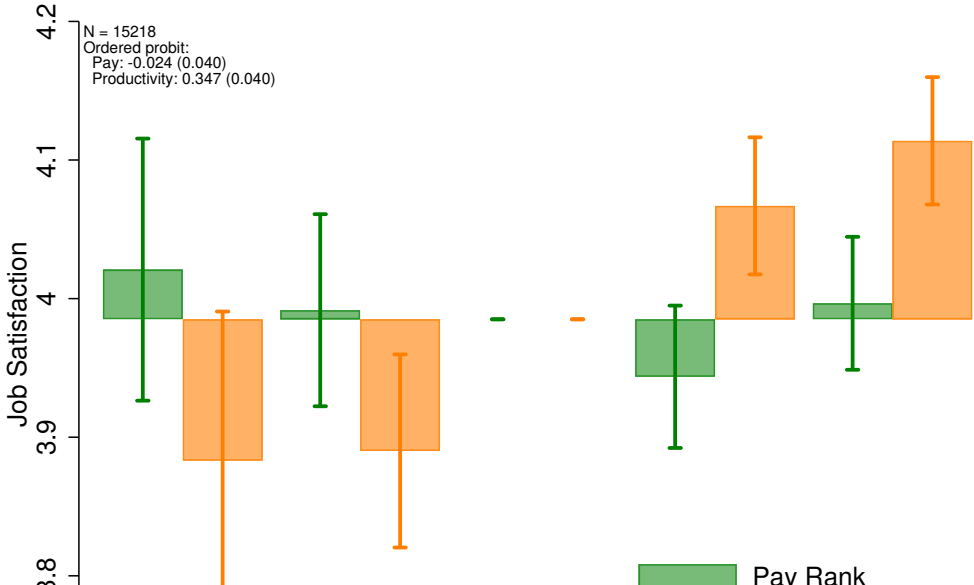
Validation: Offer Acceptance in Finland



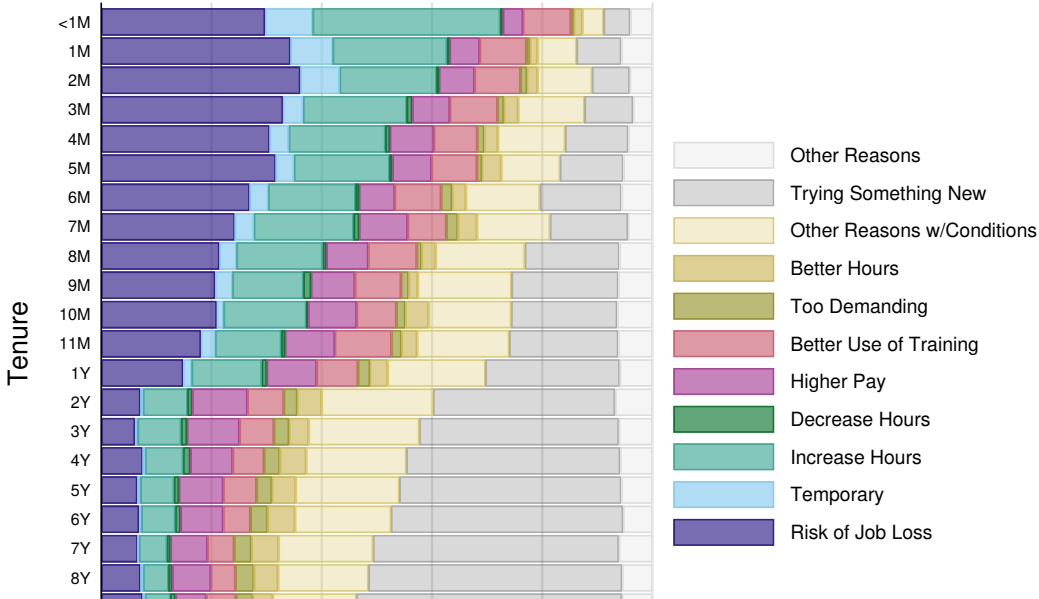
Validation: Incumbency Premium



Productivity more strongly predicts job satisfaction



Motive composition similar to firm-based job ladder



Selection into Search and Recruitment

	Offer acceptance (%)	Recruitment share (%)	Recruitment elasticity
Employed searchers	**67.2**	**24.9**	**2.33**

Selection into Search and Recruitment

	Offer acceptance (%)	Recruitment share (%)	Recruitment elasticity
Employed searchers	67.2	24.9	2.33
Unemployed searchers	**94.9**	**52.3**	**1.16**

Selection into Search and Recruitment

	Offer acceptance (%)	Recruitment share (%)	Recruitment elasticity
Employed searchers	67.2	24.9	2.33
Unemployed searchers	94.9	52.3	1.16
All searchers	**83.8**	**77.2**	**1.53**

Selection into Search and Recruitment

	Offer acceptance (%)	Recruitment share (%)	Recruitment elasticity
Employed searchers	67.2	24.9	2.33
Unemployed searchers	94.9	52.3	1.16
All searchers	83.8	77.2	1.53
Nonsearchers	**25.3**	**22.8**	**3.93**

Selection into Search and Recruitment

	Offer acceptance (%)	Recruitment share (%)	Recruitment elasticity
Employed searchers	67.2	24.9	2.33
Unemployed searchers	94.9	52.3	1.16
All searchers	83.8	77.2	1.53
Nonsearchers	25.3	22.8	3.93
Total recruitment	**54.9**	**100.0**	**2.08**

- Restricting to searchers **understates firm's recruitment elasticity by ~26%**.

Selection into Search and Recruitment

	Offer acceptance (%)	Recruitment share (%)	Recruitment elasticity
Employed searchers	67.2	24.9	2.33
Unemployed searchers	94.9	52.3	1.16
All searchers	83.8	77.2	1.53
Nonsearchers	25.3	22.8	3.93
Total recruitment	54.9	100.0	2.08

- Restricting to searchers **understates firm's recruitment elasticity by ~26%.**
- **Removing resets**: 19%

Within-Spell Search Dynamics

	Pooled	Within employer spell
	Tenure gradient (pp. per year)	
Data: Any Search	**-0.152**	**0.652**
	(0.008)	(0.064)

Within-Spell Search Dynamics

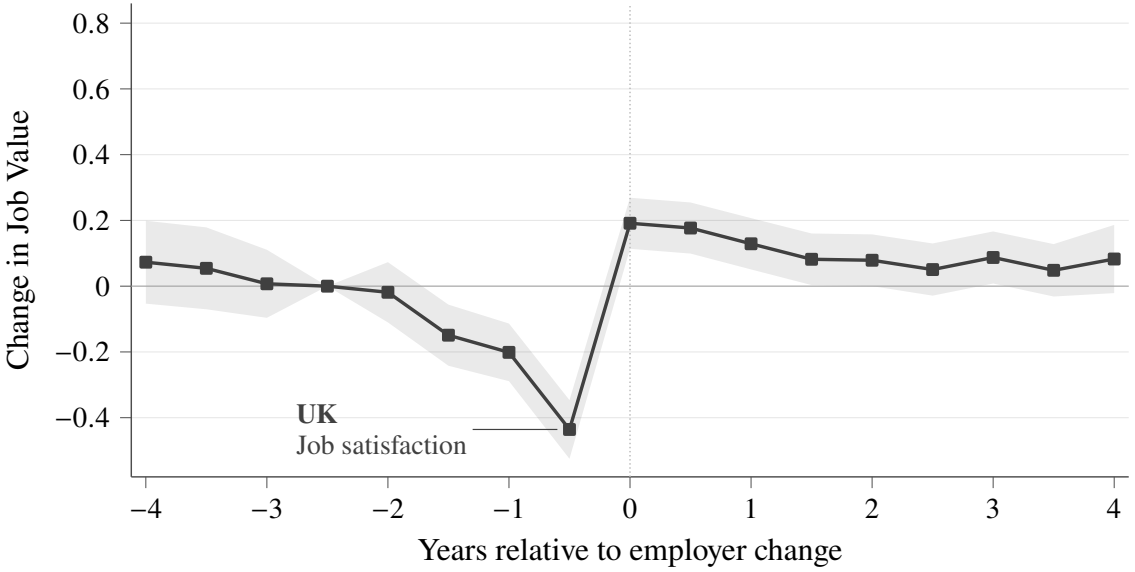
	Pooled	Within employer spell
	Tenure gradient (pp. per year)	
Data: Any Search	-0.152	0.652
	(0.008)	(0.064)
Baseline, $\zeta \approx 0.0062$	**-0.114**	**0.192**
	(0.004)	(0.041)

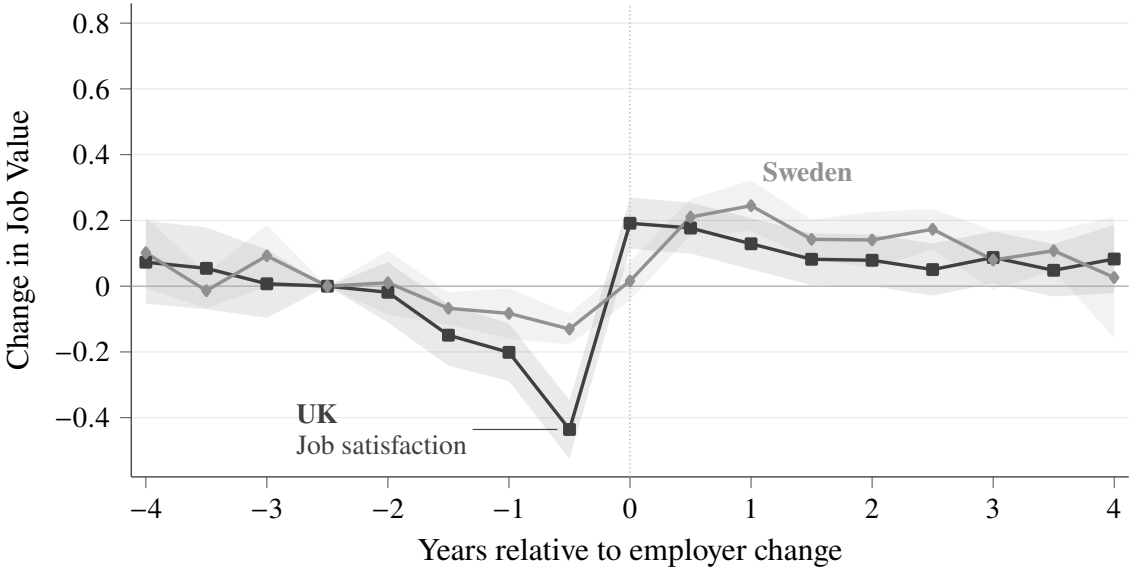
Within-Spell Search Dynamics

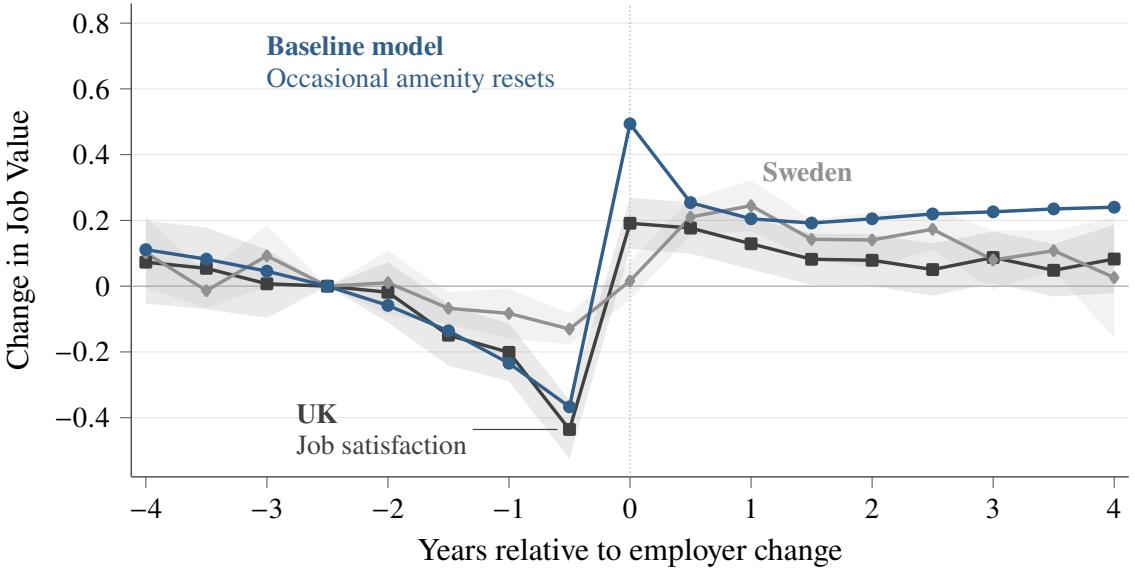
	Pooled	Within employer spell
	Tenure gradient (pp. per year)	
Data: Any Search	-0.152	0.652
	(0.008)	(0.064)
Baseline, $\zeta \approx 0.0062$	-0.114	0.192
	(0.004)	(0.041)
No resets, $\zeta = 0$	**-0.186**	**-0.109**
	(0.003)	(0.037)

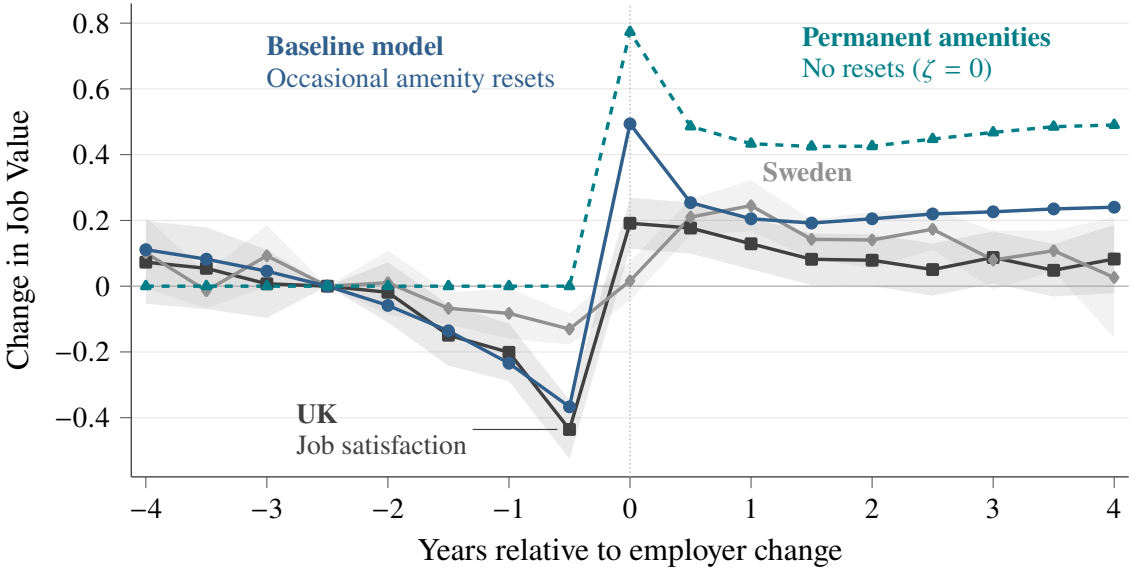
Within-Spell Search Dynamics

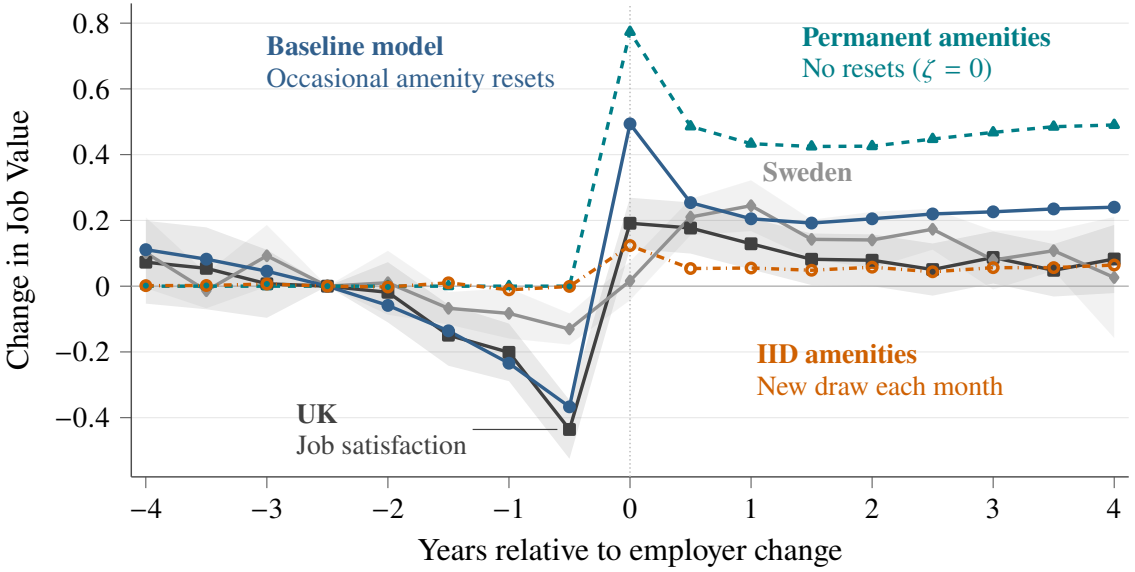
	Pooled	Within employer spell
	Tenure gradient (pp. per year)	
Data: Any Search	-0.152	0.652
	(0.008)	(0.064)
Baseline, $\zeta \approx 0.0062$	-0.114	0.192
	(0.004)	(0.041)
No resets, $\zeta = 0$	-0.186	-0.109
	(0.003)	(0.037)
IID amenities, $\zeta = 1$	**-0.015**	**-0.020**
	(0.001)	(0.014)

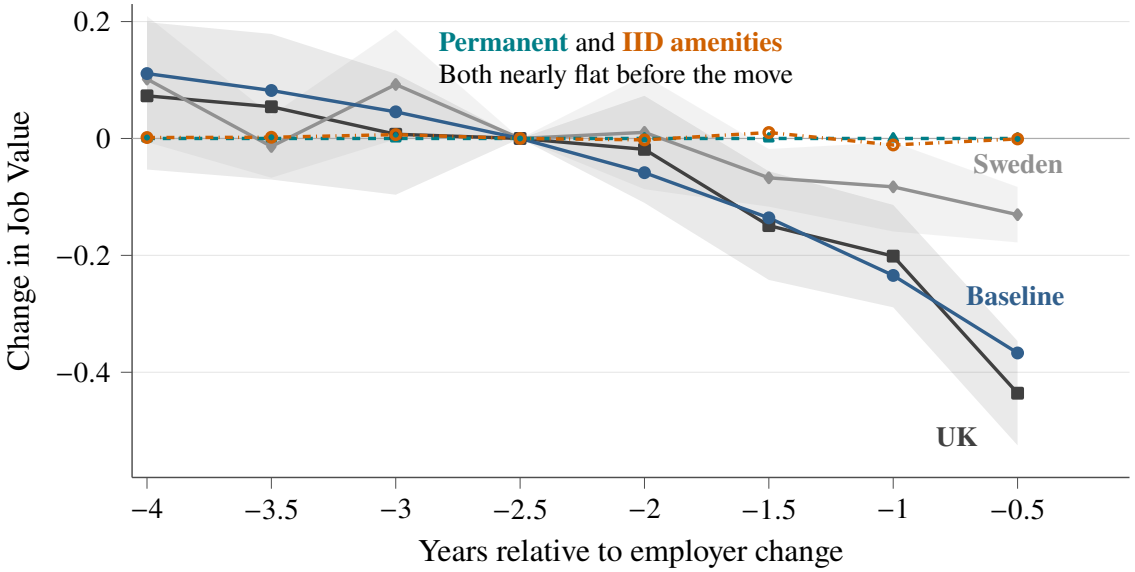

0.8
0.6
0.4
0.2
0
−0.2
−0.4
Change in Job Value
UK
Job satisfaction
−4
−3
−2
−1
0
1
2
3
4
Years relative to employer change


0.8
0.6
0.4
0.2
0
−0.2
−0.4
Change in Job Value
Sweden
UK
Job satisfaction
−4
−3
−2
−1
0
1
2
3
4
Years relative to employer change


Baseline model
Occasional amenity resets
Sweden
UK
Job satisfaction
Change in Job Value
0.8
0.6
0.4
0.2
0
−0.2
−0.4
−4
−3
−2
−1
0
1
2
3
4
Years relative to employer change


Baseline model
Occasional amenity resets
Permanent amenities
No resets ($\zeta = 0$)
Sweden
UK
Job satisfaction
Change in Job Value
0.8
0.6
0.4
0.2
0
−0.2
−0.4
−4
−3
−2
−1
0
1
2
3
4
Years relative to employer change


Baseline model
Occasional amenity resets
Permanent amenities
No resets ($\zeta = 0$)
Sweden
IID amenities
New draw each month
UK
Job satisfaction
Change in Job Value
0.8
0.6
0.4
0.2
0
−0.2
−0.4
−4
−3
−2
−1
0
1
2
3
4
Years relative to employer change


Permanent and IID amenities
Both nearly flat before the move
Sweden
Baseline
UK
Change in Job Value
0.2
0
−0.2
−0.4
−4
−3.5
−3
−2.5
−2
−1.5
−1
−0.5
Years relative to employer change